



COMSATS University Islamabad, Pakistan

FOREVER MODEST

**A project presented to
COMSATS Institute of Information Technology, Islamabad**

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of the requirement for the degree of**

Bachelor of Science in Computer Science (2021-2025)

By

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It is to certify that the final year project of BS (CS) “**Forever Modest**” was developed by **FATIMA KANWAL (CIIT/FA21-BCS-039)** a n d **M I N A L SAEED (CIIT/FA21-BCS-046)** under the supervision of “**DR. MUHAMMAD SHOAIB**” and co supervisor “**MR. MUHAMMAD JAMIL**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

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Executive Summary

In an era where digital shopping dominates the retail landscape, customers demand more than just a product—they seek complete experience that includes convenience, security, support, and speed. Forever Modest is an e-commerce platform created to fulfill this demand by offering seamless and reliable online shopping experience, with a focus on modest fashion, lifestyle products, and customer satisfaction.

Forever Modest offers a wide range of products including clothing, jewelry, and accessories. The website provides services such as **24-hour delivery**, **free shipping**, and **easy returns**. Built using modern web technologies like **Laravel** (PHP framework) for backend development, **MySQL** for database management, **HTML/CSS/Bootstrap** for responsive frontend design, and **JavaScript** for interactivity, the platform ensures performance, scalability, and user engagement.

The project introduces a **secure, code-based payment system**, which allows customers to complete purchases without needing credit or debit cards, making it ideal for customers in regions with limited online banking access. Additionally, a **simple AI-based chatbot** is integrated into the **Contact and Support** section to handle common customer queries efficiently. This AI component helps reduce the response load on human support agents and enhances the overall user experience by answering frequently asked questions and guiding users in real-time.

The admin panel provides functionalities for managing products, orders, customers, and returns. It also supports future expansion, such as integrating third-party APIs for shipping, payment gateways, and product recommendations using AI.

Overall, Forever Modest aims to be a complete digital shopping solution that blends modern design, intelligent features, and robust functionality to serve customers with efficiency and elegance.

Acknowledgement

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Abbreviations

SRS	Software Requirement Specification
PC	Personal Computer
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
MVC	Model View Controller
API	Application Programming Interface
URL	Uniform Resource Locator

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Chapter 1

Introduction

1. Introduction

This chapter provides an in-depth introduction to the final year project titled "**Forever Modest**", which is an e-commerce web application built to serve a wide range of users by offering modest clothing, stylish accessories, footwear, and elegant jewelry through a single platform. The chapter outlines the background, objectives, methodology, relevance to academic coursework, reviewed literature, challenges encountered, and the rationale for the tools and techniques selected for the system's development.

"Forever Modest" is designed to be a user-centric, aesthetically pleasing, and functionally rich online shopping platform that eliminates the dependency on physical store presence. The idea stems from the necessity to overcome market limitations and customer reach that traditional retail stores experience. The project's goal is to offer a centralized, always-accessible portal for fashion products, where users can not only view and purchase items, but also benefit from supportive features like live customer service, efficient order tracking, secure checkout, and a smooth return policy.

1.1. Project and Brief

Forever Modest is an online shopping platform developed to enhance customer convenience and extend business outreach beyond the limitations of a physical store. This project focuses on building a responsive and user-friendly website that includes features such as product listings, customer feedback, order tracking, AI-driven customer support, and code-based payment options. The technologies used in the development include **Laravel**, **MySQL**, **HTML**, **CSS**, and **JavaScript**. For email verification, the system utilizes **SMTP protocol**, eliminating the need for additional third-party APIs like Twilio or Google Maps.

The development process follows the **Incremental Software Development Lifecycle (SDLC) model**, enabling modular, iterative progress with continuous testing and integration. This report details the system's architecture, development workflow, and the implementation of intelligent customer support components.

1.2. Relevance to Course Modules

This project showcases practical applications of various theoretical and technical modules studied during the academic program. The key subjects that directly relate to this project include:

- **Web Engineering:** Used to build the front-end of the platform using HTML5, CSS3, Bootstrap, and JavaScript for responsive, mobile-friendly design.
- **Software Engineering:** Adopted the Incremental Software Development Lifecycle (SDLC) model for structured and progressive development.
- **Database Systems:** Implemented a relational database using MySQL to manage user details, product information, and order transactions.
- **Artificial Intelligence:** A basic AI-based chatbot was conceptualized for customer support, although its full integration could not be completed.
- **Human-Computer Interaction (HCI):** Principles were applied to ensure smooth navigation, clear layout design, and a responsive user interface.
- **Networking & Communication:** APIs and server-side protocols were utilized for handling order confirmations, email notifications, and dynamic data transfer.

1.3.Scope of the Project

The scope of *Forever Modest* encompasses the complete design and development of an e-commerce web platform that caters to the general public, offering a diverse range of products including modest clothing, accessories, shoes, jewelry, and more. The platform is built to support smooth and secure shopping experience with functionalities such as product browsing, user registration and login, shopping cart management, order placement, email-based order confirmation, and customer support

The system is developed solely with a **user panel**, omitting an external admin dashboard interface; however, all admin functionalities like managing product data and handling orders are performed through the backend database. The platform is hosted locally using the **XAMPP server**, and the core technologies used include **Laravel (PHP framework)** for backend development, **MySQL** for the database, and **HTML, CSS, JavaScript, and Bootstrap** for the frontend interface.

The application is intended to support

1. A responsive and visually appealing layout accessible across different devices.
2. A simple checkout process with **Cash on Delivery (COD)** as the primary payment option.
3. Email confirmation of orders using **SMTP**
4. Features like 24-hour delivery, easy returns, and customer support

The scope **does not** include live payment gateway integration or a fully functional chatbot system. While initial plans for these modules were considered, constraints related to time and complexity led to their exclusion. However, placeholders and basic design mockups have been created for future scalability. This ensures that the platform can be extended with minimal structural changes.

1.4. Motivation

The motivation for developing *Forever Modest* stemmed from the rising importance of digital presence in today's commerce landscape. In an era where online shopping rapidly overtaking traditional retail, especially post-pandemic, small and medium-scale businesses without online platforms face limitations in terms of reach and customer engagement.

Forever Modest, originally a small local business, lacked an online storefront and was restricted to physical customer interactions. This created a gap in its accessibility, particularly for users who prefer to browse and shop online. This project seeks to bridge that gap by offering a complete e-commerce solution tailored to the local context—simple, reliable, and aligned with cultural values related to modest fashion.

Furthermore, the demand for platforms with category-rich structures—especially for products targeting both men and women—further motivated the creation of a well-organized, intuitive web solution. The project also aligns with academic learning, providing practical experience in applying software engineering models, database design, interface development, and user-centered design principles.

1.5.Aims and Objectives

The main aim of this project is to design and implement a fully functional e-commerce website that is:

- User-friendly
- Visually appealing
- Technically robust
- Scalable and secure

To achieve this aim, the project pursues the following key objectives:

1. **Develop a dynamic product catalog** that categorizes items into clothing, jewelry, accessories, shoes, and gender-specific filters for men, women, and kids.
2. **Implement user authentication and account features** including registration, login, and profile management using Laravel's built-in authentication system.
3. **Enable shopping cart and order management functionality**, allowing users to select, update, and confirm product purchases through an intuitive interface.
4. **Use email confirmation via SMTP protocol** to ensure order validation and enhance reliability.
5. **Design a clean, responsive front-end interface** using HTML5, CSS3, Bootstrap, and JavaScript, ensuring accessibility across mobile and desktop devices
6. **Structure the backend using Laravel and MySQL**, maintaining organized code and secure database operations for product handling, order records, and user data.
7. **Incorporate customer service elements**, including return and refund policies, 24-hour delivery labels, and a basic support section.
8. **Apply the Incremental Software Development Model** to implement modules

iteratively, ensuring testing and refinement at each stage of development.

9. **Prepare for future scalability** by including placeholders for AI chatbot support and payment integration systems.

1.6. Project Background

"Forever Modest" originated from the observed limitations of physical stores and the need to move towards a digital platform that allows for increased accessibility and reach. In the modern world, customers demand convenience, variety, and speed. Traditional shops lack 24/7 access, quick browsing, or search capabilities. This project aims to resolve such issues by creating a centralized web-based platform, where users can shop from anywhere and anytime.

The project's background is also tied to local trends in e-commerce. Many small and medium enterprises in Pakistan still do not maintain an online presence. By bringing "Forever Modest" online, it becomes possible to tap into a much larger user base and elevate the brand to compete at the national level. The system is designed to accommodate future scalability and growth, should more categories, features, or product lines be added later.

1.7. Literature Review

To better understand the current state of the e-commerce ecosystem in Pakistan, a review of existing platforms was conducted. Several existing sites were analyzed in terms of product diversity, interface usability, category management, and feature richness.

For instance, **Prislo.com** is known to focus heavily on cosmetic and electronic products, lacking variety in fashion or modest wear. Its interface does not provide advanced category filtering, which makes it hard for users to browse specific product types. **Khazanay.pk**, on the other hand, presents a limited range of accessories with minimal category organization for gender-specific shopping. Lastly, **Metro.pk**, although well known in the grocery space, does not extend into fashion or jewelry categories, making it unsuitable for users seeking personal style products.

These observations demonstrated a major gap in the market—especially for platforms that support category-rich browsing for men, women, and children across modest fashion lines. "Forever Modest" attempts to address this void by offering a well-structured category system, intuitive interface, and features like order tracking, contact support, and responsive design.

- **Prislo.com** primarily focuses on cosmetics and technological products. It lacks a comprehensive category system, especially in fashion and accessories. Users searching for diverse product types like clothing, shoes, or jewelry find the site limiting.
- **Khazanay.pk** has a minimal representation of accessories and lacks an organized structure to differentiate items for men and women. This limits personalized shopping and makes navigation difficult for users seeking gender-specific products.
- **Metro.pk**, while known for groceries and household items, does not provide a proper fashion or accessories category. Its catalog for clothes and personal style products is underdeveloped, making it unsuitable for customers seeking complete wardrobe solutions.

1.8. Analysis from Literature Review

Based on the literature findings, several improvements were built into "Forever Modest" to provide superior user experience:

- The platform offers extensive **product categorization** for all user types (men, women, kids).
- It includes **gender-based filtering**, making it easier to find specific clothing or accessories.
- The design is **responsive and device-compatible**, allowing users to access it through phones or desktops.
- A **chatbot** was conceptualized to provide AI-driven assistance, although full deployment was not achieved within this version.
- **Order confirmation via email**, code-based payment systems, and **24-hour support policy** have been added to improve customer trust and convenience.

These enhancements ensure that Forever Modest not only competes with but also surpasses the features offered by current local e-commerce platforms.

1.9. Methodology and Software Lifecycle for This Project

The **Incremental Model** was selected as the most appropriate software development methodology for this project. It breaks down the project into manageable pieces and allows progressive building and testing of modules. Initial phases included the implementation of the user interface, navigation, and login functionality. Later increments added product listings, cart and checkout features, and database integration.

This model provided opportunities to gather feedback at each stage, ensuring continuous improvement. It also made it easier to test modules in isolation before final integration, thus reducing risk. Since the system includes several interconnected modules such as authentication, product display, and checkout, the incremental approach allowed for smooth handling of dependencies and updates.

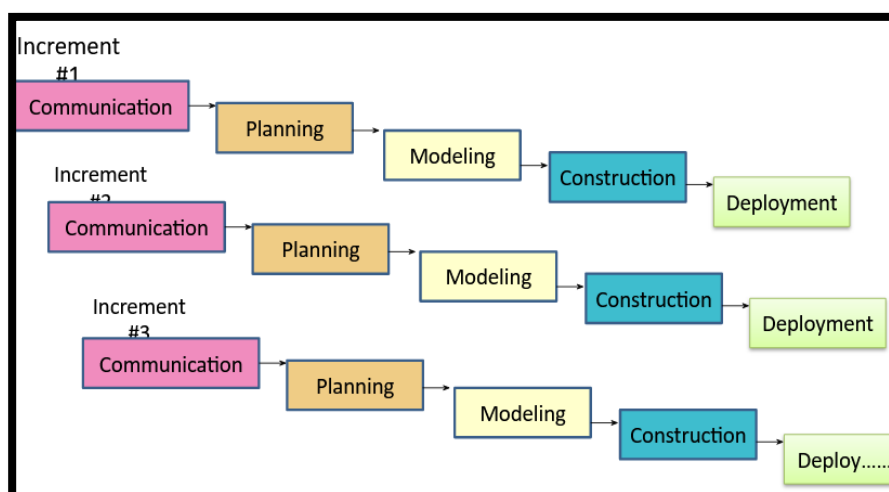


Figure 1.1 Incremental Model Diagram

1.10. Rational behind Selected Methodology

The rational for selecting the Incremental Model lies in its ability to accommodate evolving project requirements. While developing a real-time e-commerce platform, continuous enhancements are common as users interact with the system and suggest improvements. This model supported such flexibility. Additionally, it allowed clear separation of frontend and backend components, and simplified the management of module dependencies.

Had a more rigid model such as Waterfall been chosen, mid-project adjustments like chat integration or design layout changes would have been difficult. The Incremental approach facilitated easier testing, improved debugging, and allowed the system to be deployed in stages, which aligned perfectly with our development and evaluation goals.

1.11. Challenges Faced During Development

While the project was completed with core features intact, several challenges surfaced during the development process. One of the main limitations was the **incomplete integration of chatbot functionality**, which had been planned using AI but was excluded due to time and scope constraints. Similarly, **payment gateway integration** using third-party services like Stripe or JazzCash was planned but could not be implemented due to restricted access and API documentation issues.

Another challenge was ensuring **database synchronization and query optimization**, especially as product and user data increased. Managing data relationships efficiently while maintaining speed was essential to avoid lag during user browsing or cart actions. Additionally, **designing a consistent UI across various screen sizes** was initially problematic, but later resolved through Bootstrap's grid system and media queries.

Lastly, handling **user authentication, session control**, and admin-level functions from the database without a dedicated admin panel interface required careful planning and precise implementation using Laravel's MVC architecture.

Chapter 2

Problem Definition

2. Problem Definition

This chapter delves into the precise problem that **Forever Modest** addresses, outlines the deliverables and developmental prerequisites of the project, and examines the existing offline processes that the new platform will replace. It also presents a thorough requirements analysis—covering both functional and non-functional requirements—and detailed use case descriptions to guide design and implementation

2.1. Problem Statement

Forever Modest currently operates without an online interface, managing all transactions and customer interactions physically within its store. This manual approach results in limited scalability, inefficiencies in processing customer orders, and an inability to capture or analyze data effectively. In contrast, competitors with robust e-commerce solutions offer 24/7 accessibility, personalized recommendations, and seamless checkout experiences. Without a modern digital platform, Forever Modest cannot compete effectively or meet evolving customer expectations.

To overcome these challenges, the project will develop a fully functional e-commerce website that supports secure user registration, product browsing, shopping cart management, and order fulfillment. The system will be built using an incremental development model that allows core functionality to be delivered early and refined over successive iterations.

2.2. Deliverable and Development Requirements

The following items constitute the key deliverables of the project. These outputs will ensure that the software meets functional, non-functional, and user requirements. The **Incremental Model** was ideal for this project because it allows flexibility and early user feedback. Since e-commerce systems involve several interdependent modules (e.g., login, cart, payment, order tracking), this model helped ensure each module was tested thoroughly before moving on to the next. Moreover, this approach made it easier to accommodate new features like AI support and third-party API integration without redesigning the entire system.

The success of this project hinges on the delivery of the following key components:

1. **E-commerce Frontend:** A responsive web interface enabling customers to browse product categories, view item details, and manage their shopping carts.
2. **User Authentication Module:** Secure registration and login system with password hashing, session management, and account recovery via SMTP-based email verification.
3. **Shopping Cart & Checkout:** A robust cart mechanism that persists between sessions, applies discount codes, and supports Cash-on-Delivery (COD) payment method, followed by an email confirmation upon order placement.
4. **Order Management Backend:** A database-driven workflow for processing orders, updating stock levels, and recording transaction histories.
5. **Admin Functions (Database Tools):** Although no graphical admin dashboard

is provided, administrators can manage product and order records directly via MySQL or phpMyAdmin interfaces.

6. **Support & Feedback Module:** A contact form integrated with SMTP for customer inquiries and feedback storage in the database.

The development process will follow the **Incremental SDLC model**, allowing each module to be designed, implemented, and tested in isolation before integration. Early iterations will focus on the core shopping flow (product listing → cart → order confirmation), while later increments will introduce support modules and refinement of the user interface.

2.3.Deliverables

- Fully functional e-commerce website
- Secure customer login & registration system
- Shopping cart and checkout system with online payment integration
- Admin panel for product, category, order, and user management
- Optional chatbot for basic customer service
- Contact page with Google Maps location
- Order history

2.4.Technical Development Requirements

The following table, Table 2.1, presents the technologies and tools used in the development of the Forever Modest e-commerce platform. This style of table numbering and captioning will be followed consistently throughout the report.

Table 2.1: Technologies and Tools Used in the Development of Forever Modest

Component	Technology/Tool Used
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Laravel (PHP framework)
Database	MySQL
Server	XAMPP
Development a Platform	Visual Studio Code
Design Tools	Figma,

2.5.Current System

The current system is entirely offline and manually managed. Customers visit the physical store to browse products, place orders, and make payments. Inventory and customer data are maintained using spreadsheets and paper records. There is limited reach beyond the local customer base, and the absence of an e-commerce platform restricts growth.

The existing workflow at Forever Modest is entirely manual:

- **Customer Interaction:** Shoppers visit the physical store to browse items, request assistance, and place orders over the counter.
- **Order Recording:** Staff record each sale and customer detail on paper order forms.
- **Inventory Management:** Stock levels are tracked in spreadsheets maintained offline; updates occur only when staff manually reconcile sales data.
- **Order Fulfillment:** Orders are prepared and handed to customers; there is no automated notification or tracking.
- **Customer Feedback:** Feedback is collected verbally or via paper forms, then filed for later review.

The following figure, **Figure 2.1**, represents the **Current System Workflow of Forever Modest**, providing an overview of the existing manual and offline processes, including in-store customer visits, product browsing, order placement, and inventory management using paper records and spreadsheets.

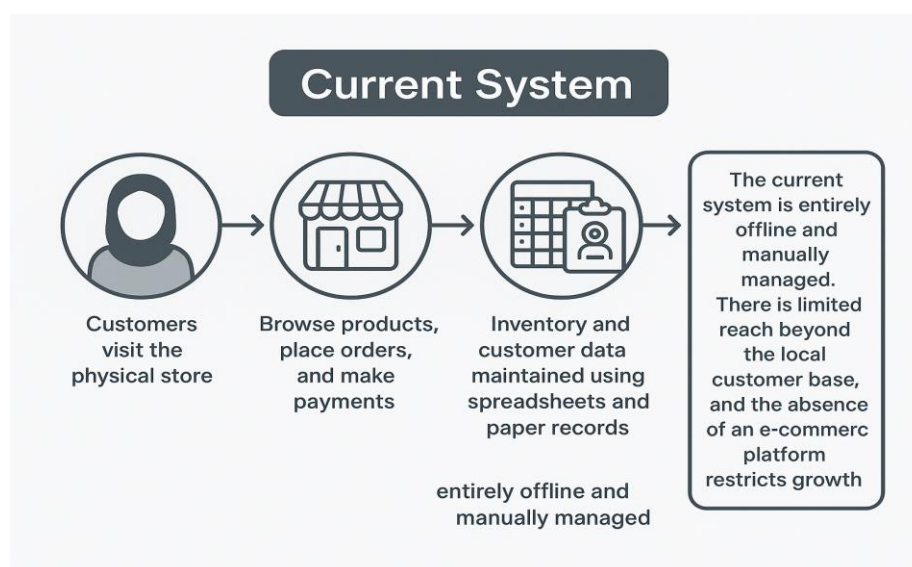


Figure 2.1: Current System Workflow of Forever Modest

This manual model introduces time delays, transcription errors, and limited reporting capabilities. Transitioning to an online platform will automate data capture at each stage, enabling faster processing, accurate reporting, and improved customer communication.

Chapter 3

Requirement Analysis

3. Requirement Analysis

Requirement Analysis is a crucial phase in system development, where the goal is to understand the needs and expectations of the stakeholders and to translate them into a clear set of system requirements. For the **Forever Modest** e-commerce platform, this process involves analyzing both the functional and non-functional requirements, as well as identifying the key user interactions through **use case diagrams**.

3.1. Use Case Diagram

A **Use Case Diagram** serves as a high-level visual representation of how users (or actors) interact with the system. For the **Forever Modest** e-commerce platform, the diagram includes two primary actors: **Customer** and **Admin**. These actors interact with various system functionalities to perform key tasks. **Customer**: A Customer can browse products, add them to the cart, place orders, and view their order history. They can also register, log in, and manage their profile.

Admin: An Admin can manage product listings, process orders, manage customer accounts, and oversee system operations

These interactions are represented through various **use cases** such as **Login, View Products, Add to Cart, Place Order**, and more

The following **figure 3.1** illustrates the use case diagram for the "Forever Modest" e-commerce platform. It visualizes the interaction between two primary actors: the **Customer** and the **Admin**. The diagram highlights the key functionalities available to each actor.

A use case diagram provides a visual summary of user interactions with the system. Key actors include **Customer** and **Administrator** (via database tools). Primary use cases are:

- Register / Login
- Browse Products
- Add to Cart / Update Cart
- Checkout & Place Order
- Receive Confirmation Email
- Submit Inquiry / Feedback
- Manage Products (Admin)
- Manage Orders (Admin)

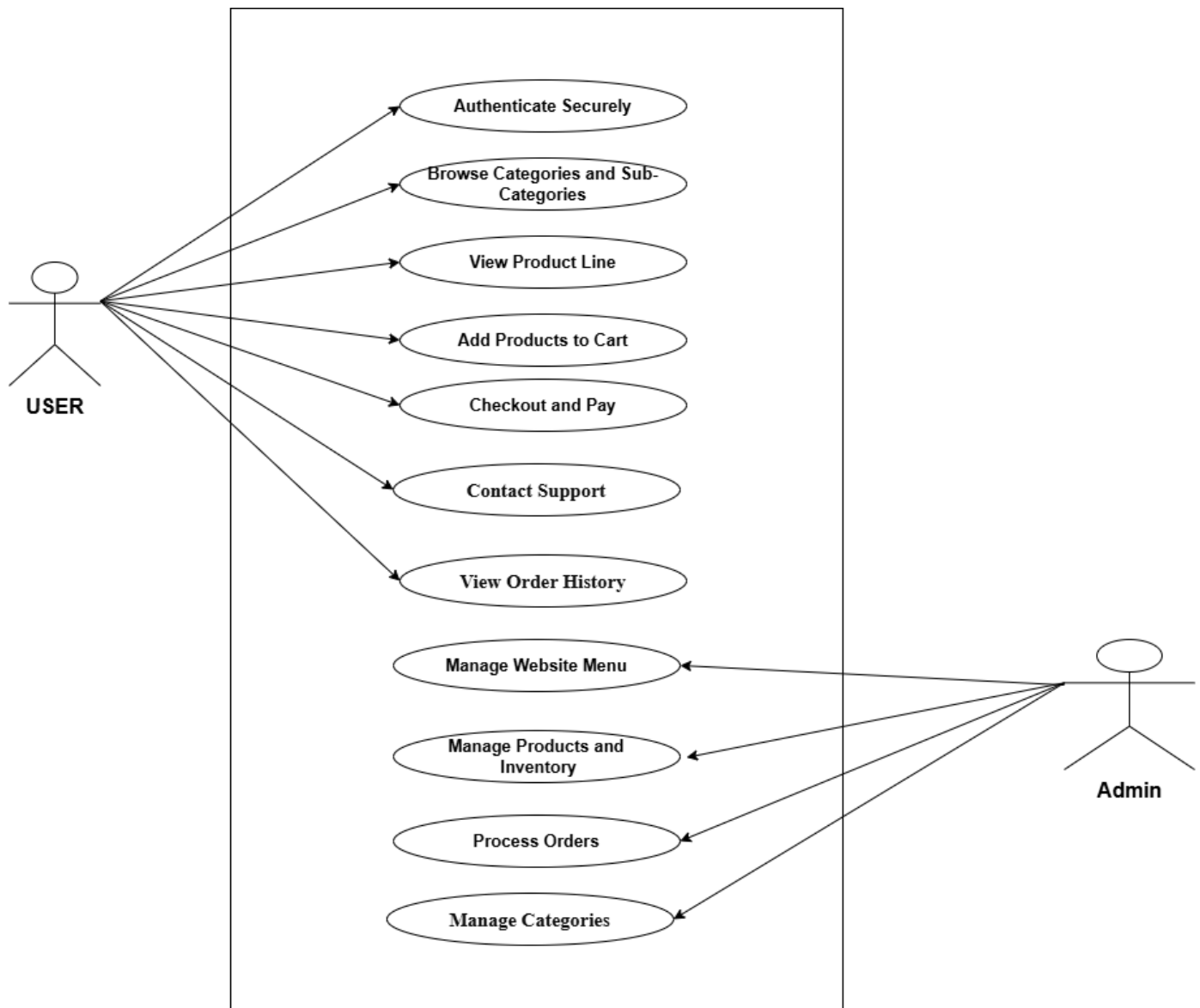


Figure 3. 1 Use Case Diagram for Forever Modest E-Commerce Platform

3.2. Use Case Descriptions

Below is a table describing the key use cases identified for the **Forever Modest** platform. Each use case includes the involved actors, description, and the flow of events.

The following table illustrates the use case scenarios for the "Forever Modest" e-commerce platform, categorized into two main sections.

Use Case Name: Customer and Admin roles

Actor: Customer and Admin

Precondition: The customer is registered and logged in. Products must have been added to the cart. and the admin is managing the product

3.2.1. Main Flow:

1. The system displays order summary with item list and total price.
2. The customer confirms shipping details and selects COD payment.
3. The system validates stock availability for each item.
4. The system creates a new order record in the database.
5. The System decrements product stock quantities.
6. The System sends a confirmation email via SMTP.
7. The system displays an order confirmation page with order ID.

3.2.2. Customer Use Cases

Customer Use Cases outline the primary actions available to customers, such as browsing products, placing orders, managing accounts, updating personal details, and tracking order status.

These cases ensure a smooth and user-friendly shopping experience

Table 3.1: Customer Use Case Details for "Forever Modest" E-Commerce Platform

Use Case	Description
Browse Products	Customers can view and search through the product catalog.
View Shopping Cart	Customers can see the items added to their cart.
Place Order	Customers can complete the order by entering shipping details and payment.
Track Order	Customers can monitor the status of their orders.
Manage Account	Customers can view or update their personal and payment details.

3.2.3. Admin Use Cases:

This table outlines the use cases for the administrative side of the "Forever Modest" platform. It includes tasks such as product management, order processing, customer support, content updates, and overseeing payment systems, ensuring smooth and efficient platform operations.

Table 3.2 Admin Use Case Details for "Forever Modest" E-Commerce Platform

Use Case	Description
Add/Update/Delete Products	Admins can manage the product catalog, including adding, updating, or removing products.
Manage Orders	Admins can process and update orders based on their status.
Process Payments	Admins handle payment verification and processing.
Handling Customer Queries	Admins respond to and resolve customer service inquiries.

3.3. Functional Requirements

Functional requirements specify **what** the system must do to satisfy user needs and support business processes. For *Forever Modest*, these requirements break down the core features and interactions that a customer or administrator will perform. They include user registration and secure login, browsing and filtering products by category and gender, managing a persistent shopping cart (add, update, remove items), processing orders with cash-on-delivery, and sending confirmation emails via SMTP. Additional functions cover customer feedback submission through a contact form, and backend operations—performed via database tools—allowing administrators to add, edit, or remove products and orders. Each functional requirement maps directly to a use case diagram, ensuring that every user interaction (e.g., “Checkout,” “Submit Inquiry,” “View Order History”) is clearly defined, testable, and traceable through the development process.

1. **FR1 – User Registration:** The system must allow new users to register with a unique email and password.
2. **FR2 – User Authentication:** The system must authenticate returning users securely, with session management.
3. **FR3 – Product Catalog:** The system must display products by category with images, descriptions, and prices.
4. **FR4 – Shopping Cart:** The system must enable users to add, modify, or remove

items from their cart.

5. **FR5 – Checkout Process:** The system must process an order using COD and record transaction details.
6. **FR6 – Email Notifications:** The system must send an order confirmation email via SMTP.
7. **FR7 – Feedback Submission:** The system must allow customers to submit inquiries or feedback through a form.
8. **FR8 – Admin Management:** Authorized personnel must be able to add, edit, or delete products and orders using database tools.

3.4.Non-Functional Requirements

Non-functional requirements describe **how** the system performs rather than **what** it does, focusing on quality attributes that affect user satisfaction and system maintainability. For this project, performance requirements ensure page load times remain under two seconds even under moderate load, preserving a seamless shopping experience. Usability targets specify an intuitive interface requiring no more than three clicks for major tasks, with responsive design across desktop and mobile devices. Security measures mandate HTTPS for all data transmission, hashed password storage, and protection against common web threats. Scalability requirements guarantee that new features—such as online payment gateways or AI chatbots—can be integrated without major architectural changes. Finally, maintainability and compatibility stipulations ensure the codebase follows Laravel best practices, is well-documented, and supports modern browsers like Chrome, Firefox, and Edge.

1. **NFR1 – Performance:** The system must respond to page requests within two seconds for up to 500 concurrent users.
2. **NFR2 – Usability:** The interface should be intuitive, with common tasks requiring no more than three clicks.
3. **NFR3 – Security:** All user passwords must be hashed; sensitive data must be transmitted over HTTPS.
4. **NFR4 – Scalability:** The architecture must support future integration of payment gateways and chatbots without major redesign.
5. **NFR5 – Compatibility:** The system must function correctly on modern browsers (Chrome, Firefox, Edge) and mobile devices.
6. **NFR6 – Maintainability:** Code should follow Laravel best practices, with clear documentation and modular structure.

Chapter 4

Design and Architecture

4. Design and Architecture

This chapter outlines the core system architecture, data representation, process flow, and design models of the *Forever Modest* e-commerce platform. These design elements ensure the system is scalable, maintainable, and fully aligned with the functional and non-functional requirements defined in Chapter 2.

4.1. System Architecture

The overall architecture of Forever Modest employs a classical three-tier model, separating concerns into Presentation, Business Logic, and Data layers. This separation enhances maintainability, fosters independent development, and enables future scalability for features such as AI-based services.

4.1.1. Presentation Layer (Web Frontend)

- Implements responsive user interfaces using **HTML5**, **CSS3**, **Bootstrap**, and **JavaScript**.
- Adopts mobile-first design principles, ensuring usability on various screen sizes through CSS media queries and Bootstrap's grid system.
- Utilizes client-side scripting for interactive components—such as dynamic product filtering and cart updates—without full page reloads.

4.1.2. Business Logic Layer (Application Server)

- Developed on the **Laravel** framework, following the **Model-View-Controller (MVC)** pattern to organize code into clear, testable modules.
- Controllers handle HTTP requests, orchestrate workflows (e.g., checkout, login), and invoke model methods for data operations.
- Models represent domain entities (User, Product, Order), encapsulate database interactions via Laravel's Eloquent ORM.
- Middleware components enforce security policies, such as **authentication**, **authorization**, and **input validation**, guarding against unauthorized access and common attacks.

4.1.3. Data Layer (Database Server)

- Built on **MySQL**, hosted via **XAMPP** for local development and testing.
- Employs **InnoDB** storage engine to ensure ACID-compliant transactions and foreign-key support.
- Database schema is normalized to Third Normal Form (3NF) to eliminate redundancy and maintain data integrity.

- Data access is optimized with indexes on frequently queried columns (e.g., `product_id`, `user_id`) and careful query structuring.

4.1.4. Communication Flow

- The client issues HTTP(S) requests to the web server, which are routed to appropriate controllers.
- Controllers interact with Eloquent models, executing CRUD operations.
- Responses are rendered as HTML views or JSON (for AJAX calls), updating the frontend.
- Email notifications—such as order confirmations—are sent via SMTP through Laravel’s mail service, decoupling email logic from core business code.

The structure of the system explains the working of the system and describes the components of the project, their relationships, and how they interact with each other and with the architecture. As Shown in Figure 4.1.

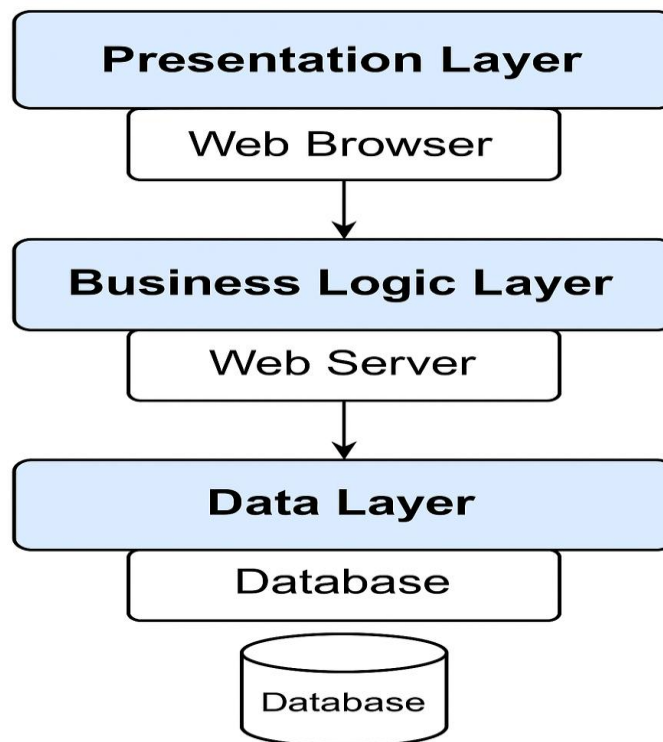


Figure 4.1 System Architecture Diagram of Forever Modest E-Commerce Website

4.2. Data Representation

The platform’s data model captures essential business entities and their relationships. A well-

structured **Entity-Relationship Diagram (ERD)** defines cardinalities and foreign-key constraints, ensuring reliable data storage and retrieval.

The data representation follows a relational database model, with entities representing core business objects:

- **User:** Stores information like user ID, name, email, password, and address.
- **Product:** Contains product ID, name, description, price, image, and stock.
- **Order:** Records order ID, user ID, product list, total amount, status, and timestamps.
- **Cart:** Tracks items temporarily added by the user for purchase.
- **Payment:** Holds transaction details including payment method, date, and amount.

4.2.1. Key Entities and Attributes

1. **Customer:** customer_id (PK), name, email, password_hash, address, phone, created_at, updated_at.
2. **Product:** product_id (PK), name, category, description, price, stock_quantity, image_url, created_at, updated_at.
3. **Order:** order_id (PK), customer_id (FK), total_amount, status, order_date, shipping_address.
4. **Order_Item:** order_item_id (PK), order_id (FK), product_id (FK), quantity, unit_price.
5. **Cart_Item:** cart_item_id (PK), customer_id (FK), product_id (FK), quantity, added_at.
6. **Payment:** payment_id (PK), order_id (FK), payment_method, amount, payment_date.

4.2.2. Relationships and Cardinalities

1. A **Customer** can place **many Orders** (1:N).
2. An **Order** comprises **multiple Order Items** (1:N), each linking to a single **Product** (N:1).

3. A **Customer** can have **many Cart Items** (1:N) before checkout.
4. Each **Order** has one **Payment** record (1:1) under the current COD model.

Normalization to 3NF ensures that each non-key attribute depends solely on its primary key, preventing update anomalies. Indexes on foreign-key columns speed up join operations between tables.

Each entity has well-defined attributes and relationships (one-to-many and many-to-many) to ensure normalization and data integrity across the system.

This diagram illustrates the relationships between key entities such as Users, Orders, Products, and Payments. It helps visualize how data is structured and connected within the system's database.

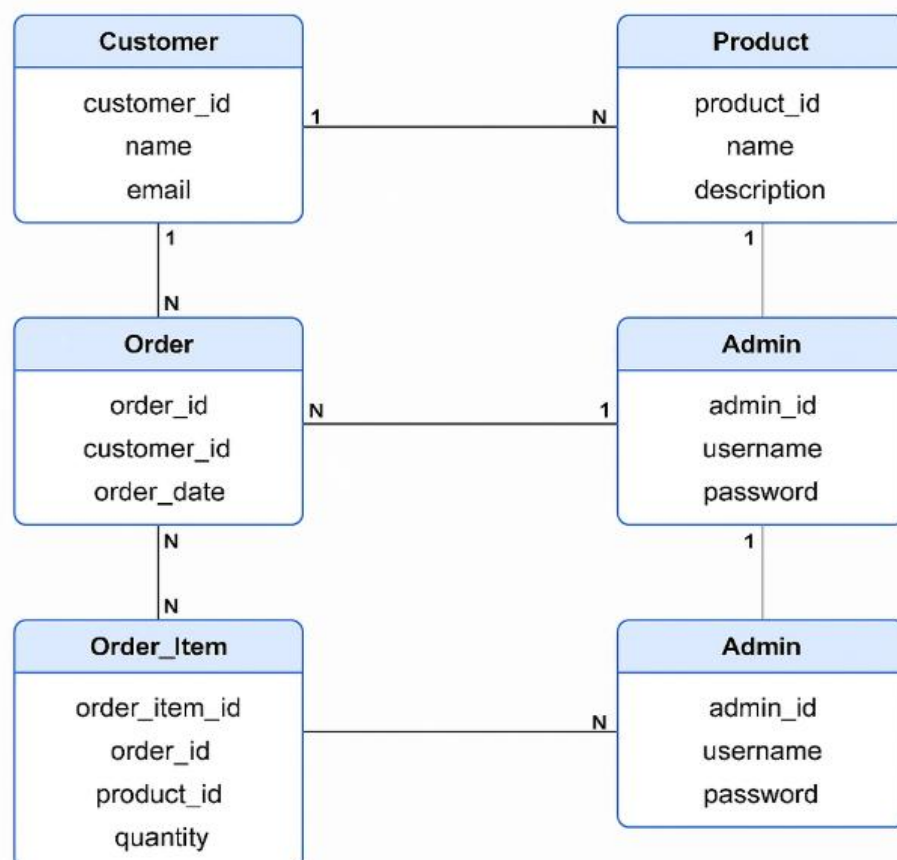


Figure 4.2 Entity Relationship Diagram (ERD) for the Forever Modest E-commerce System.

4.3. Process Flow/Representation.

This section describes the high-level process flow for key user interactions, emphasizing the

sequence of steps and backend triggers.

This process flow diagram depicts the step-by-step interaction from the user's perspective when placing an order on the platform:

- **Login/Register:** The user authenticates or creates a new account.
- **Browse Products:** The user views categories, filters, and selects items.
- **Add to Cart:** Selected items are added to the cart.
- **Checkout:** The user reviews the cart and proceeds to place an order.
- **Payment:** Transaction is securely processed.
- **Order Confirmation:** An order ID is generated, and the inventory is updated.

The backend services are triggered appropriately to manage database updates, payment verification, and product stock handling.

4.3.1. User Order Process Flow

1. **Authentication:** User visits the site and either registers a new account or logs in. Credentials are verified, and a session is established.
2. **Product Browsing:** User navigates through categories, applies filters (e.g., price range, gender), and views detailed product pages.
3. **Cart Management:** When a user clicks "Add to Cart," an AJAX call updates the Cart_Item table. The cart icon dynamically reflects the item count without full page reload.
4. **Checkout Initiation:** User reviews the cart, enters or confirms shipping details, and selects the Cash-on-Delivery option.
5. **Order Processing:** Upon checkout, a database transaction creates an Order record and associated Orde_Item entries. The system decrements stock_quantity for each purchased product.
6. **Payment & Confirmation:** Order details are recorded; a payment record is created under COD. The user receives an email confirmation via SMTP, and the frontend displays an order summary.
7. **Post-Order Management:** Administrators monitor orders through phpMyAdmin, update order status as "Processing" or "Shipped," and customers can later view their order history.

This flow ensures atomic operations; if any step fails (e.g., low stock), the transaction rolls back, and the user is notified to adjust their cart.

This diagram shows the main steps users follow on the Forever Modest platform, from browsing products to completing an order.

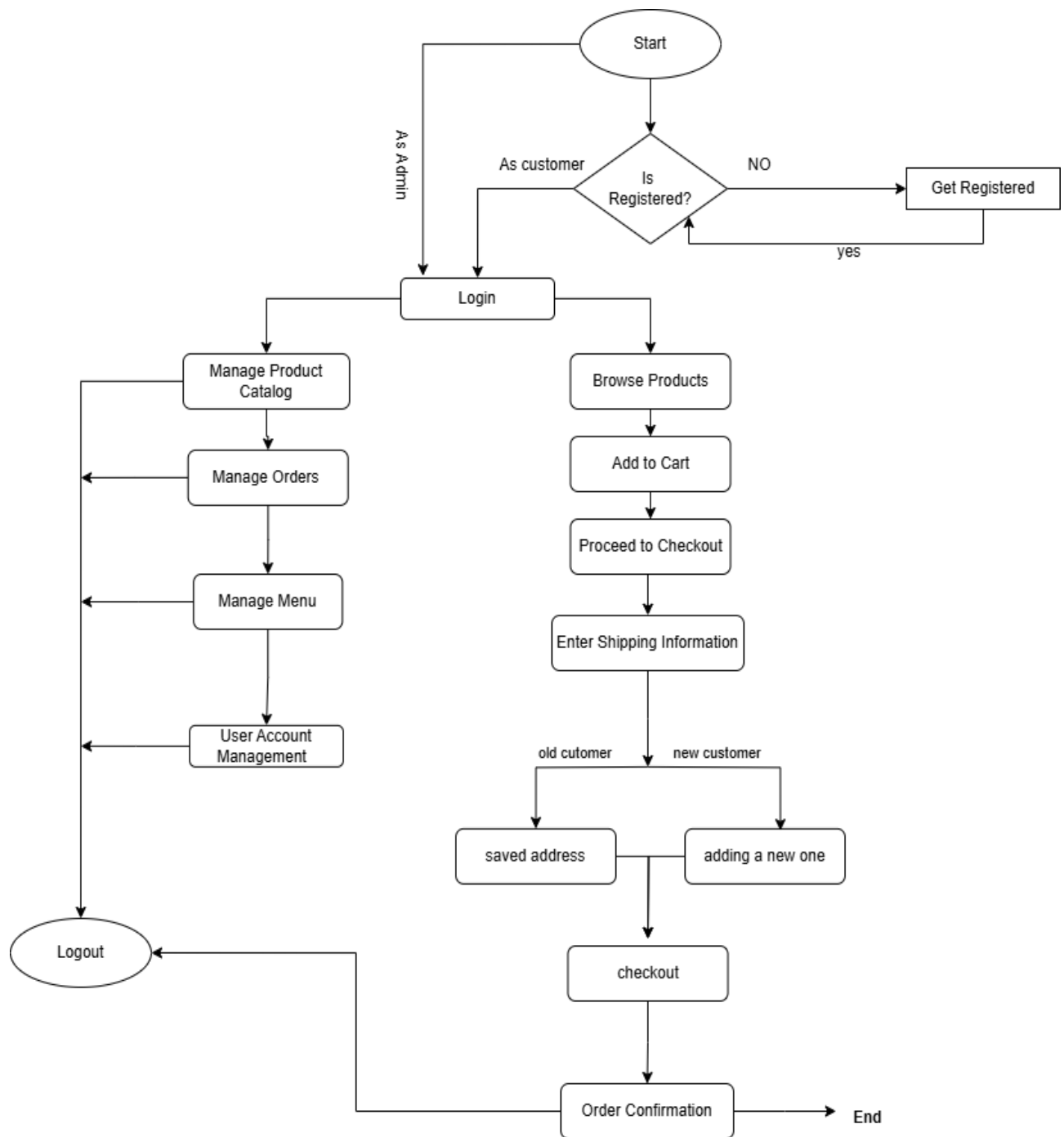


Figure 4.3 User Order Process Flow

4.4. System Modules

The system is organized into a collection of specialized modules, each dedicated to a specific domain of functionality and encapsulating related responsibilities. This modular approach promotes clear separation of concerns, making it easier to manage code complexity by isolating features into well-defined components. Each module can be developed, tested, and deployed independently, facilitating parallel development efforts and reducing inter-module dependencies. Modularity also simplifies future enhancements and maintenance tasks, as improvements or fixes in one area have minimal impact on other parts of the system, thus ensuring greater stability and scalability.

1. **Authentication Module:** Handles user registration, login, logout, password hashing, and session management. It ensures secure access to the application and enforces role-based permissions.
2. **Product Module:** Manages product catalog operations, including adding, editing, categorizing, and displaying products. It interacts with the database to fetch product details and supports filtering and search capabilities.
3. **Cart Module:** Enables users to add, update, and remove items from their shopping cart. It maintains cart state in the database across sessions and calculates subtotal amounts dynamically.
4. **Order Module:** Orchestrates the checkout workflow. It validates cart contents, creates order and order-item records, updates inventory levels, and handles order status transitions.
5. **Payment Module:** Implements the Cash-on-Delivery (COD) mechanism. Although no third-party payment gateway is integrated, this module records payment intent, links payments to orders, and ensures the correct flow of financial data.
6. **Email Notification Module:** Utilizes SMTP to send automated emails for account verification, order confirmations, and support ticket updates. It decouples email logic from core business workflows using Laravel's mail services.
7. **Support Module:** Provides a contact form for user inquiries and feedback. It captures messages in the database and triggers email notifications to administrators for prompt follow-up.
8. **Administration Module:** Although lacking graphical UI, administrative tasks such as product and order management are performed through direct database tools (phpMyAdmin). This module defines database schemas and stored procedures to support bulk operations and reporting.

Chapter 5

Design Models

5. Design Models

The design models describe the structural and behavioral aspects of the system. These models serve as blueprints for developers and provide a clear view of how different components interact with each other. The design models provide both structural and behavioral perspectives of the Forever Modest system, serving as detailed guides for developers, testers, and stakeholders. These models help in visualizing interactions over time and the static organization of classes and components. Four primary sequence diagrams are presented to capture distinct user and admin interactions, followed by a comprehensive class diagram illustrating the core object-oriented architecture

Below are the primary models used

5.1. Sequence Diagram

This sequence diagram shows the interaction between the frontend, user service, order service, inventory service, and payment gateway during the order placement process. The flow includes:

- Request initiation from user interface
- Validation by the user service
- Order creation request to the order service
- Payment confirmation via the payment gateway
- Inventory update to reflect new stock levels
- Final confirmation and notification to the user

This ensures that operations are executed in a controlled and logical sequence.

5.1.1. Sequence Diagram: User Registration Flow

The **User Registration** sequence diagram outlines the interactions between a new user, the frontend interface, and backend services to create an account. It begins with the user submitting registration details and ends with successful account setup and confirmation email dispatch.

1. User fills out the registration form and clicks "Sign Up".
2. Frontend sends POST request to `AuthController@register`.
3. `AuthController` validates input and calls `UserService.createUser()`.
4. `UserService` hashes the password and persists the User record.
5. `EmailService` sends a welcome email via SMTP.
6. Frontend receives success response and redirects the user to the login page.

This diagram represents how a user registers or logs in to the e-commerce website to begin shopping.

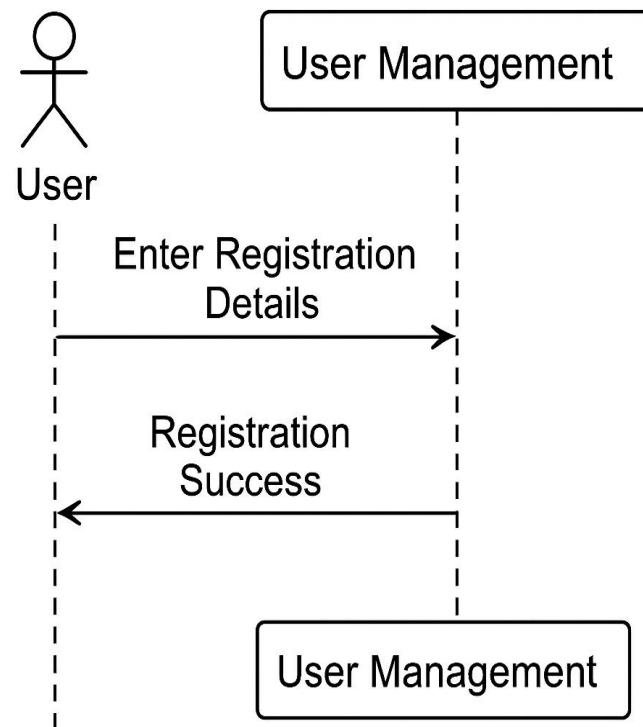


Figure 4.4 Sequence Diagram – User Registration

5.1.2. Sequence Diagram: Product Search and Detail View

The second sequence diagram focuses on how a user's search request is processed. It begins with a search input on the user interface, which triggers a call to the ProductController. The controller interacts with the ProductService to query the database, filter results, and return product details. This interaction includes input validation, search optimization, and error handling for no-results scenarios.

This diagram illustrates the interaction between the user and the product module when a user searches for a product and views its details

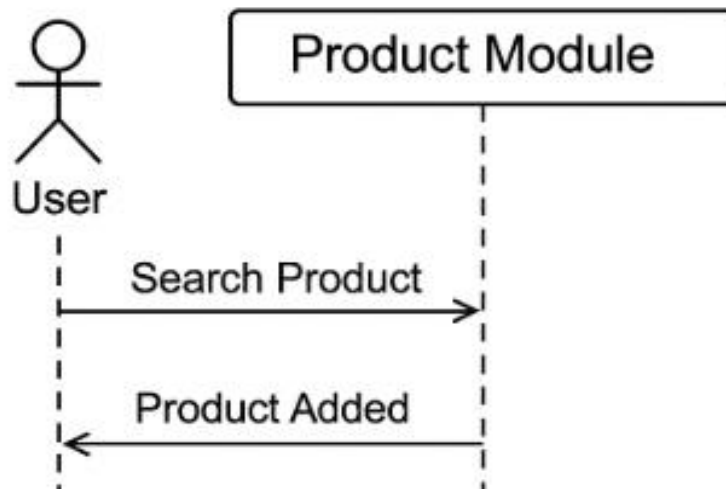


Figure 4.5 User searches and views product details

5.1.3. Sequence Diagram: Cart Management

The third sequence diagram illustrates the addition and removal of items from the shopping cart. Each action generates an asynchronous request to the CartController, which updates the Cart_Item table accordingly. The model also covers the response flow, showing how the UI updates the cart count and displays notifications for added or removed items.

This diagram outlines how the user adds and removes products from the cart, showcasing the core functionality of cart management.

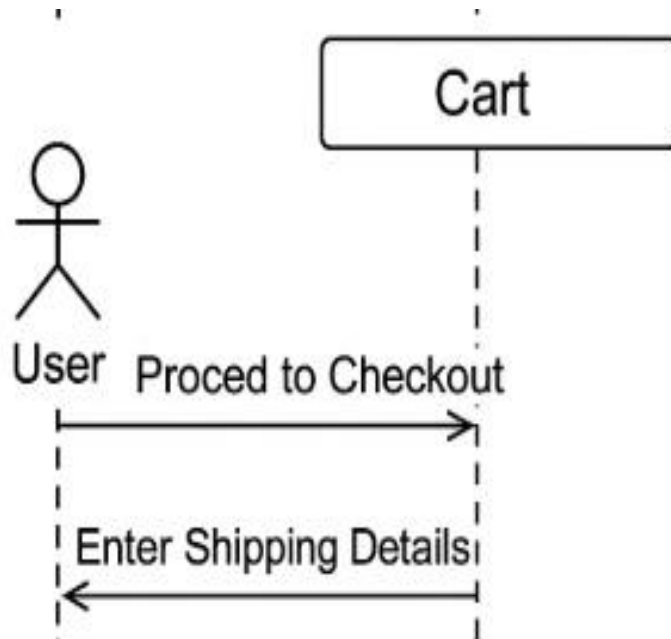


Figure 4.6 User adds or removes items from cart.

5.1.4. Sequence Diagram: Checkout with COD

This sequence diagram presents the specific pathway for a Cash-on-Delivery (COD) transaction. It shows how the CheckoutController coordinates customer data, shipping address capture, and order finalization. It also outlines exception paths, such as handling invalid addresses or canceled orders, ensuring robustness in the checkout module.

The diagram represents the checkout process where a user proceeds to check out, enters shipping details, and confirms the order. It includes an option for Cash on Delivery (COD) as the payment method.

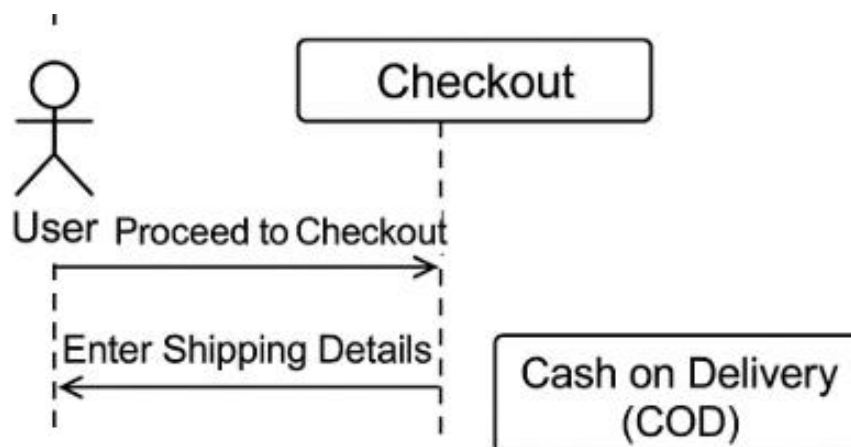


Figure 4.7 User completes checkout with COD option.

5.1.5. Sequence Diagram: Admin Product Management

The final sequence diagram captures the admin side interaction for adding or updating products. Although admin operations are performed via phpMyAdmin, this model abstracts the workflow: an admin submits product details → Backend validates input → Database schemas are updated → Confirmation is returned.

This diagram shows how the admin adds a new product or updates existing product details in the Forever Modest system.

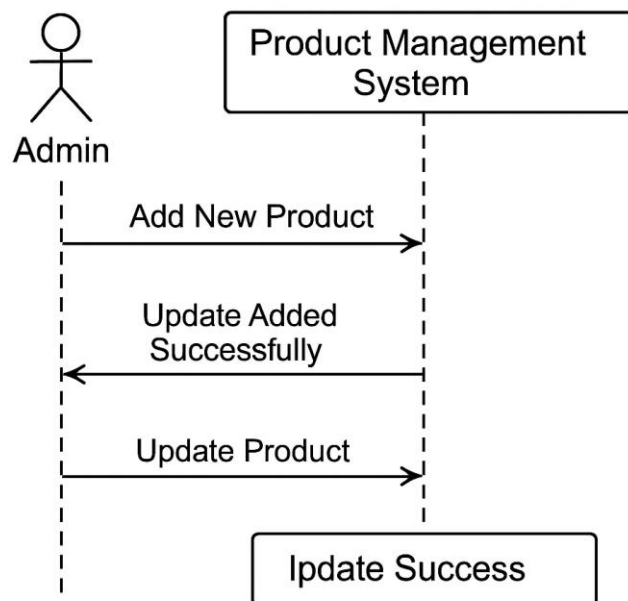


Figure 4.8 Admin adds or updates product information

5.2. Class Diagram

The class diagram offers a static overview of all major classes and their relationships. Key classes include Customer, Product, CartItem, Order, OrderItem, Payment, and EmailService. Associations such as composition (Order → OrderItem) and dependencies (CheckoutController → PaymentModule) are clearly marked, along with attributes and primary methods for each class. This diagram underpins the system's object-oriented design and guides developers in implementing class hierarchies and interfaces.

The class diagram represents the object-oriented model of Forever Modest, displaying the key classes and their interactions

- **User:** Handles personal details and user roles.
- **Product:** Contains product-specific details such as name, ID, and pricing.
- **Order:** Associates users with their purchases and tracks delivery status.
- **Cart:** Temporarily stores select items before purchase.
- **Payment:** Records and validate transaction details.

The diagram illustrates relationships such as inheritance, aggregation, and associations to define data handling in the system's codebase.

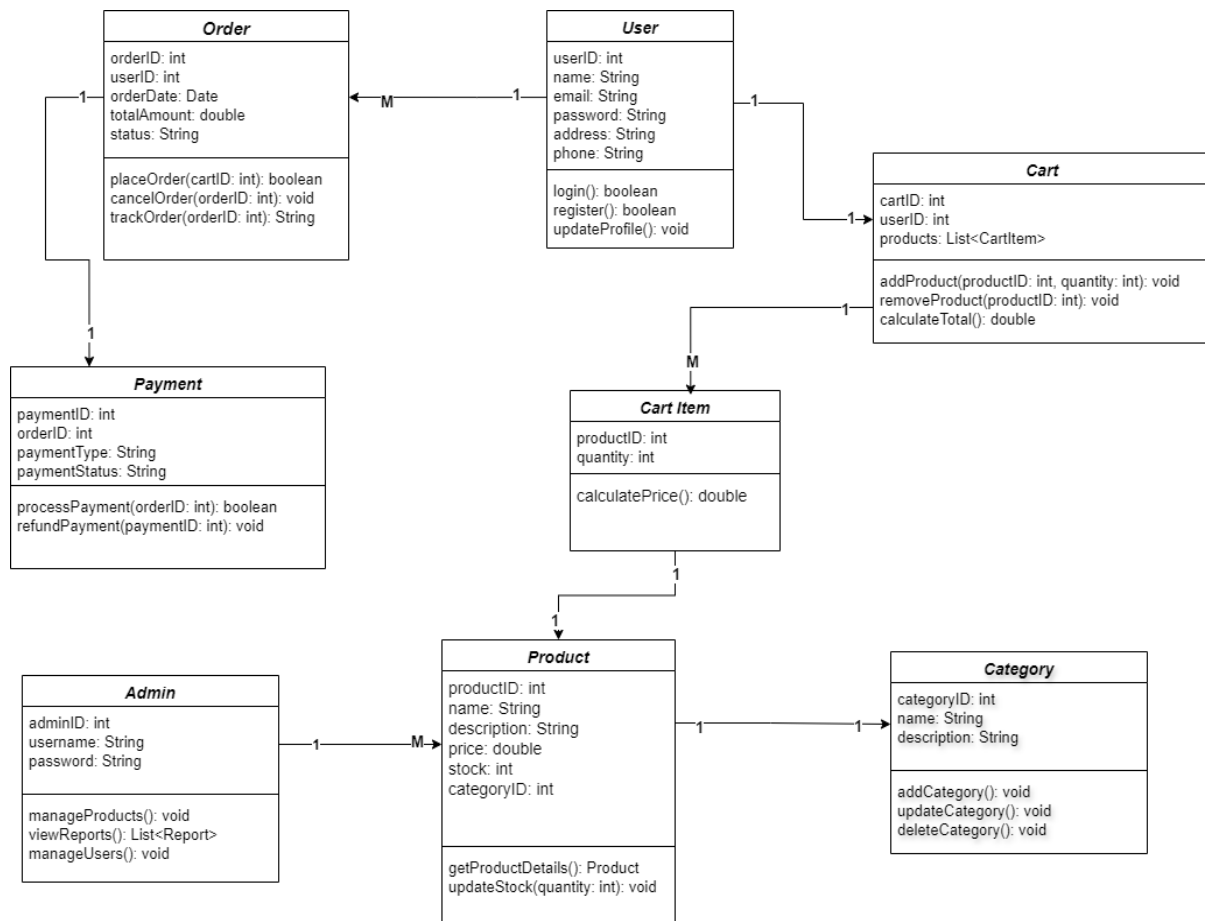


Figure 4.9 Class Diagram – System Structure

Chapter 6

Implementation

6. Implementation

This chapter will outline the implementation details of the project, describing the algorithms used, APIs integrated, and relevant functionalities. It will also reference UML diagrams for any visual representation of the system's architecture and flow.

The implementation of the **Forever Modest** e-commerce system is a culmination of thorough design, strategic planning, and the integration of intelligent components. This chapter describes how the system was practically developed, including the algorithms that govern major functionalities, the third-party APIs that extend the system's capabilities, and the design of the user interface to ensure an excellent user experience.

By covering algorithms, APIs, and UI design, this chapter bridges the gap between system architecture and actual development. Each section provides insight into the purpose, scope, and contribution of individual components to the functioning of the overall platform.

6.1. Algorithm

The system's behavior and logic are driven by foundational algorithms designed for efficiency, scalability, and a seamless user experience. These algorithms define how the system processes input, executes actions, and returns outcomes for common user operations such as authentication, product browsing, and order management.

The core functionalities of the project are built on several algorithms that drive key modules. Below is the pseudocode or natural language explanation of how the main features of the platform operate:

User Authentication Algorithm

User authentication is a critical aspect of any e-commerce system, as it governs who can access certain features of the platform. The authentication logic includes verifying user credentials, creating new user accounts, and managing user sessions.

Key Functionalities

- Secure login and signup operations.
- Backend validation of credentials using secure hashing mechanisms.
- Session management to keep users logged in securely.
- Role-based redirection (e.g., user vs. admin dashboard access).
- Login error handling and feedback for invalid credentials.
- This algorithm ensures that only authorized individuals access their data while preventing any unauthorized use of the system.

➤ Algorithm

- **Start**
- The user enters login credentials (username and password).
- Validate credentials against the database.
- If credentials are correct, grant access to the user dashboard.
- If incorrect, display an error message and prompt for re-entry.
- **End**

Product Search Algorithm

This algorithm governs how the system responds to user search queries and filter preferences. It enhances the usability and relevance of the product catalog by offering accurate and dynamic product listings

Key Functionalities:

- Real-time product search based on user input.
- Filtering by category, price, size, color, rating, or availability.
- Sorting options such as popularity, lowest price, and newest arrivals.
- Suggestive search or auto complete for a better experience.
- Handling no-result scenarios with alternative recommendations.

By making product discovery efficient and user-friendly, this algorithm significantly improves customer engagement and satisfaction.

➤ **Algorithm**

- **Start**
- The user enters a search term in the search bar.
- The system queries the database for product matches.
- Display matching products with relevant details (name, price, image).
- Sort the products based on the user's filter (e.g., price, rating).
- **End**

Order Processing Algorithm

This algorithm manages the complete lifecycle of a user's purchase—from adding items to the cart to confirm payment and finalizing the order. It integrates financial transactions, inventory updates, and order records.:

Key Functionalities:

- Collecting and summarizing cart items.
- Calculating total cost with tax and shipping.

- Offering multiple payment options (Cash on Delivery, Online Payment).
- Interacting with external payment APIs for secure transactions.
- Storing order history and generating invoice/receipt.
- Sending order confirmation emails and SMS alerts.
- This logical flow ensures accurate, reliable, and trackable transactions on the platform.

➤ **Algorithm**

- **Start**
- The customer adds products to the shopping cart.
- Calculate total price including taxes and shipping fees.
- The user enters payment details and proceeds to check out.
- Process payment using integrated API.
- Confirm payment and display order details.
- **End**

6.2. External APIs

The project integrates several external APIs to enhance functionality. These APIs handle tasks like payment processing, product recommendations, and location-based shipping. The table below provides details of each API used in the system.

6.2.1. Firebase Authentication API

Firebase Authentication simplifies user account creation and verification while maintaining high security standards. The project integrates several external APIs to enhance functionality. These APIs handle tasks like payment processing, product recommendations, and location-based shipping. The table below provides details of each API used in the system.

Purpose and Benefits:

- Simplifies secure user registration and login processes.
- Provides built-in support for email/password, phone, and social login.
- Supports multi-device session handling.
- Offers features like password reset and email verification.
- Reduces development time for secure backend logic.

- Firebase ensures that users' credentials and data are protected and managed by a globally trusted provider.

6.2.2. Cloud Storage API

Cloud storage is essential for managing static content like images and files uploaded by admins or users

Purpose and Benefits

- Enables reliable storage of product images, banners, and user avatars.
- Supports high-speed upload/download capabilities.
- Automatically scales based on the volume of media content.
- Organizes media using folders or tags for better retrieval.
- Secures image access using permission controls.

This API keeps the site visually appealing while ensuring performance.

6.2.3. Geolocation API

Purpose and Benefits

- Detects user's country, city, or ZIP code.
- Helps estimate delivery time and shipping cost.
- Enables location-specific offers or restrictions.
- Improves user experience through personalization.
- Optimizes logistics by clustering orders by region.

Integrating geolocation allows the system to better serve international users.

6.2.4. Currency Conversion API

This API helps dynamically display product prices in a currency relevant to the user

Purpose and Benefits

- Automatically detects user's preferred currency.
- Updates currency rates using real-time data.
- Converts prices across currencies with precision.
- Increases accessibility for global users.
- Reduces confusion by avoiding manual conversion.

This globalizes the platform, making it user-friendly for international shoppers.

Table 5.1 Details of APIs used in the project

Name of API	Description of API	Purpose of Usage	List of Functions/Classes Used
Firebase Authentication API	A simple API for handling user authentication, including sign-up and login features.	Used for authenticating users during login and sign-up processes.	signUpUser(), loginUser(), logoutUser()
Cloud Storage API	Provides cloud storage for storing product images, profile pictures, etc.	Used to store and retrieve product images, user profile images.	uploadImage(), getImage()
Geolocation API	A simple API to fetch the user's geolocation (location-based services).	Used to provide users with location-based shipping options.	getUserLocation(), calculateShippingCost()
Currency Conversion API	An API to convert product prices between different currencies (if applicable).	Used to display product prices in different currencies.	convertCurrency(), getExchangeRate()

6.3. User Interface

The user interface is the face of the Forever Modest platform. It must be visually appealing, intuitive, and easy to use across all devices. UI design is not only about aesthetics but also about accessibility, responsiveness, and usability.

The user interface (UI) is designed to provide a seamless shopping experience for customers. Key features include:

- A responsive design that adapts to various screen sizes and devices.
- An intuitive layout with categories, filters, and a search bar for easy navigation
- Secure and user-friendly forms to enhance trust and usability

The admin interface includes dashboards for managing inventory, tracking sales, and generating reports. Clear and concise designs ensure that both customers and administrators can interact with the platform effortlessly.

6.3.1. Home Page

The Home Page is the central hub and initial point of interaction for most users. It has been crafted to grab attention instantly while guiding users towards key areas such as product categories, ongoing promotions, or popular listings. The homepage serves as the primary entry point for users, showcasing featured products, promotions, and a visually appealing layout. It includes a navigation menu, search bar, and a banner highlighting special offers.

Main Design Features:

- **Hero Banner:** A large banner placed at the top showcasing time-limited offers or featured collections with strong call-to-action (CTA) buttons like "Shop Now" or "Explore More".
- **Navigation Bar:** Includes site-wide menu links (Home, Shop, Categories, Services, Contact), a search bar, login/signup buttons, and a shopping cart icon.
- **Search Functionality:** Allows users to input keywords to quickly find products of interest, featuring auto-suggestions.
- **Featured Products Section:** A horizontally scrollable or grid layout displaying the most popular or discounted items.
- **Visual Appeal:** Uses brand-specific color palettes, elegant typography, and high-resolution graphics for a stylish look.
- **Responsive Layout:** Fully optimized for both mobile and desktop experiences with clean, uncluttered design flow.

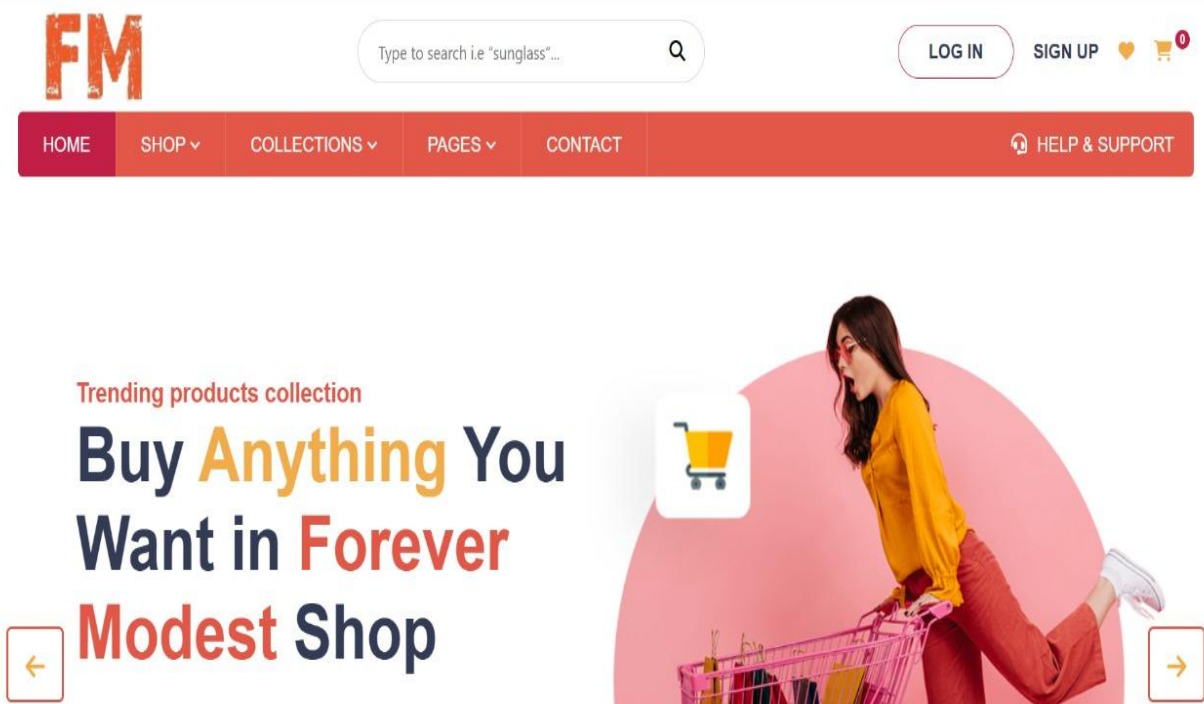


Figure 5.1 Home Page Interface Design

6.3.2. Services Page Interface design

This section outlines the services provided by Forever Modest, such as fast delivery, customer support, and hassle-free returns. The page emphasizes the brand's commitment to quality and customer satisfaction. The **Services Page** presents the brand's value-added services in a user-

friendly and informative manner. This page is essential in building trust and showcasing customer-centric commitments.

Main Design Features:

- **Service Cards/Boxes:** Each service (e.g., Fast Delivery, 24/7 Customer Support, Easy Returns) is displayed in its own dedicated section or card with an icon and short description.
- **Visual Icons:** Illustrative graphics/icons that symbolize each service for quick recognition and visual hierarchy.
- **Emphasis on Reliability:** Highlighting guarantees like “Delivery within 24 hours” or “100% Satisfaction Guaranteed”.
- **Customer-Centric Tone:** The text is written in a friendly and reassuring voice to increase user confidence.
- **User Friendly:** Cards stack neatly on smaller screens to retain readability and layout balance.

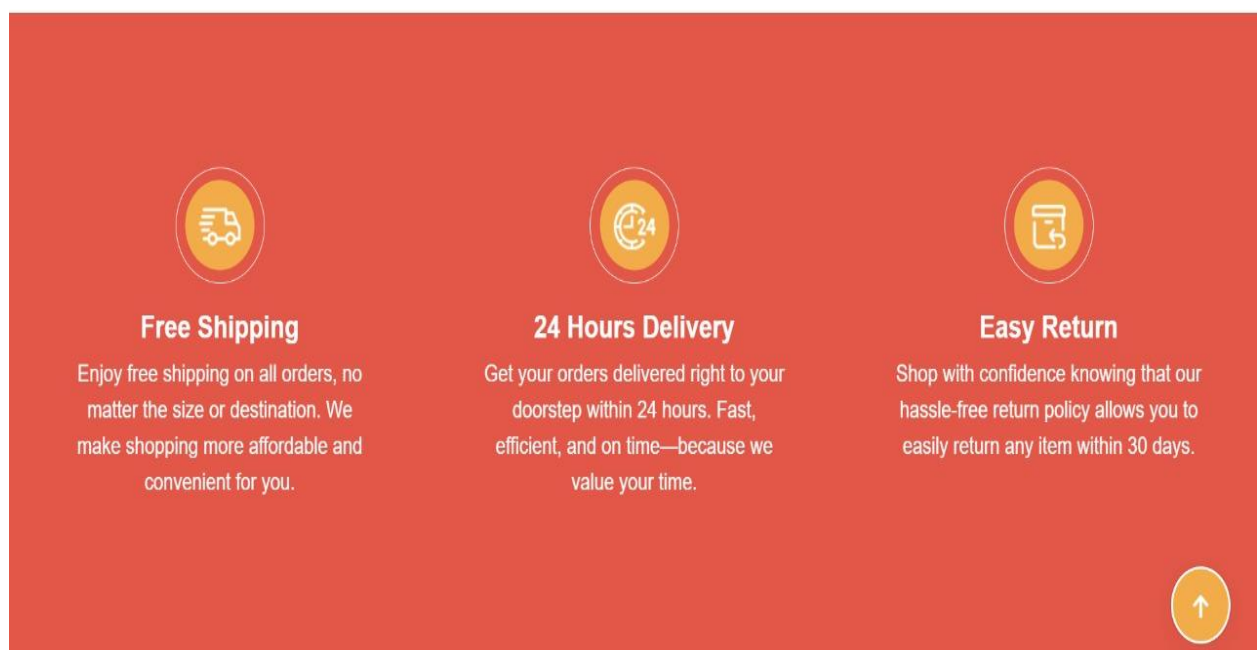


Figure 5.2 Services Page Interface design

6.3.2. Intuitive Product Categorization

Displays various product categories such as clothing, accessories, footwear, and jewelry. Each category is represented by high-quality images and clickable links that direct users

to relevant product listings.

This section illustrates how the site organizes products into intuitive categories, allowing users to find items quickly and efficiently.

Main Design Features:

- **Category Tiles:** Clearly marked tiles or image cards for Clothing, Jewelry, Accessories, Footwear, etc.
- **Clickable Navigation:** Each image or tile acts as a button, redirecting the user to the corresponding category page.
- **High-Quality Visuals:** Categories are represented by thematic, high-resolution images reflecting the items inside.
- **Consistent Layout:** Uniform sizing and spacing of tiles/cards to ensure a visually pleasing display.
- **Hover Effects:** Adds interactivity such as zoom-in or fade-in titles when users hover over a category.

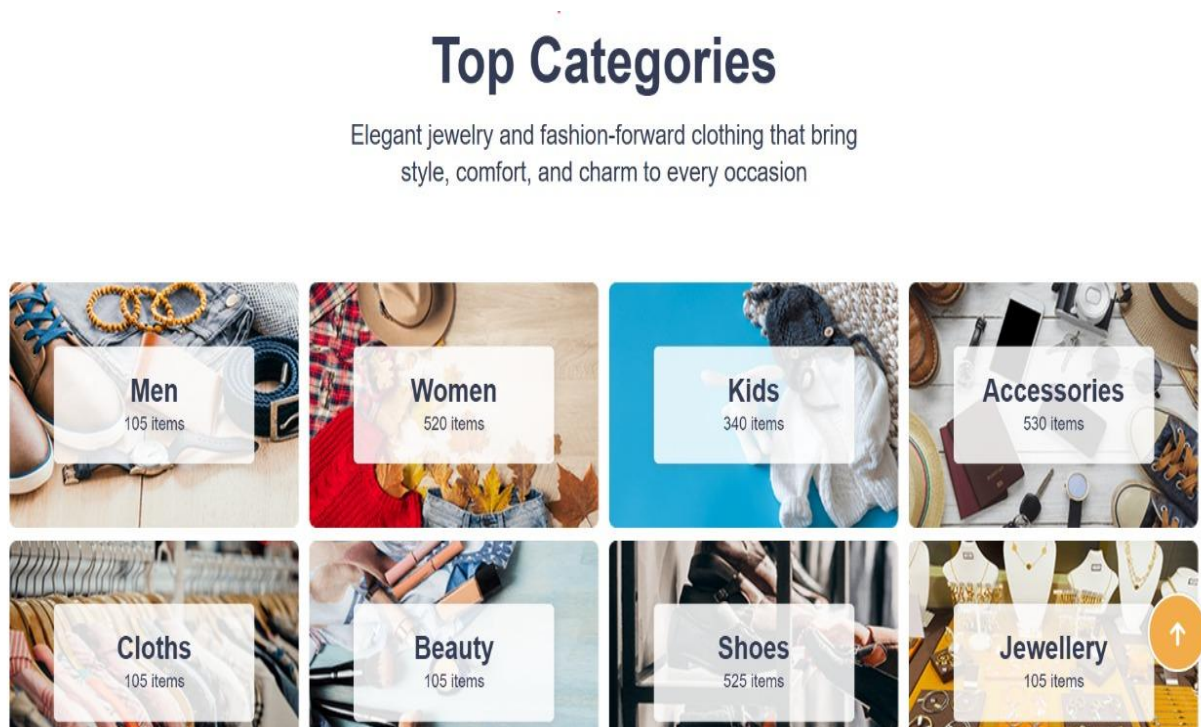


Figure 5.3 Intuitive Product Categorization

6.3.3. Detailed Item Display Interface

Showcases a grid of available products with images, names, prices, and a short description. Users can browse through items; filter based on preferences and select products for purchase. This screen allows users to browse through various products in a structured format. It presents a grid of product cards that provide essential information at a glance.

Main Design Features:

- **Product Grid:** A structured layout showcasing products in rows/columns, usually 3-4 per row on desktop and 1-2 on mobile.
- **Each Card Contains:**
 - **Product Image** – a high-quality thumbnail.
 - **Product Title** – clear, concise naming.
 - **Price Tag** – bold and eye-catching.
 - **Short Description** – optional one-liner to tease features.
- **Filters & Sort Options:** Sidebar or top panel offering options to filter by price, size, popularity, color, etc.
- **Hover Preview:** Interactive behavior showing a quick view button, favorite icon, or add-to-cart shortcut.

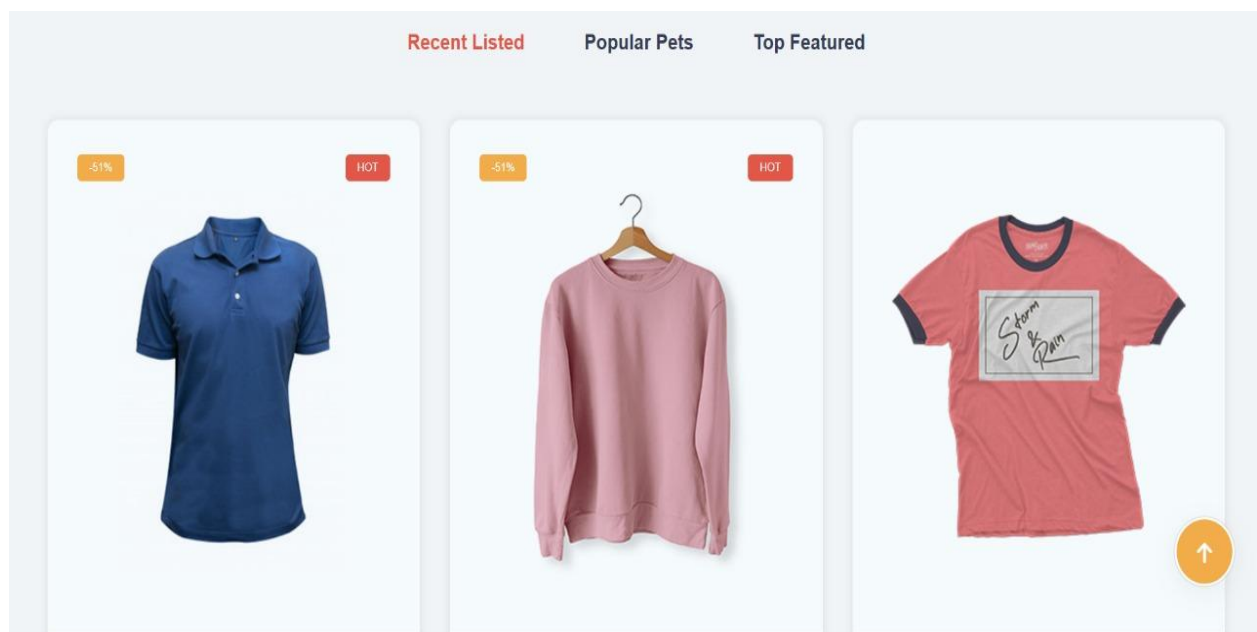


Figure 5.4 Detailed Item Display Interface

6.3.4. Highlighted Features

Highlights the key features of Forever Modest, including high-quality materials, trendy designs, and affordability.

This section reassures customers of the brand's value proposition. This section is strategically placed to emphasize the core strengths and selling points of Forever Modest. It builds trust and reinforces why customers should shop from the platform.

Main Design Features:

1. Feature Blocks

Visual boxes or sections highlighting attributes such as:

- Premium Quality Materials
- Modern and Modest Styles
- Affordable Prices
- Trusted by Thousands of Customers

- 2. Icons + Text Combo:** Each feature is paired with an appropriate icon and a brief, persuasive description.
- 3. Grid or Horizontal Layout:** Displayed in a neat and balanced fashion with good use of whitespace.
- 4. Brand Message Integration:** Each feature supports the brand's commitment to modest fashion and reliability.

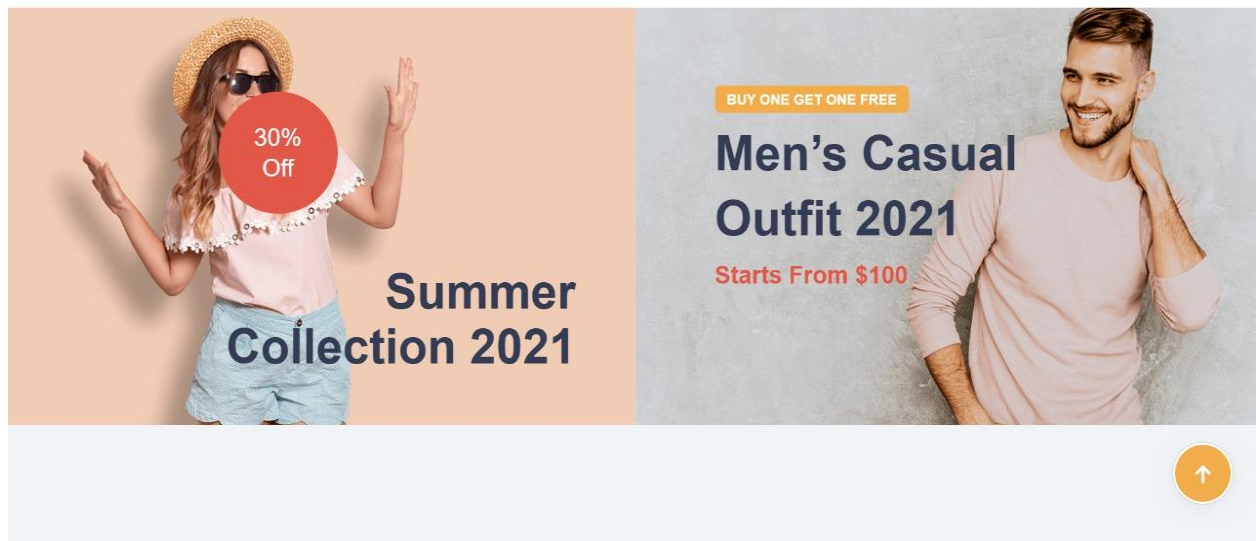


Figure 5.5 Highlighted Features Interface

6.3.5. Footer Section of the Website

Includes essential links such as About Us, Contact, Terms & Conditions, and Social Media handles. The footer ensures easy navigation to important sections of the website.

The footer acts as the bottom-most section of the website, accessible from all pages. It contains supplementary navigation and key information that users may need.

Main Design Features

- Quick Links
 - About Us
 - Contact
 - FAQs
 - Terms & Conditions
 - Privacy Policy

Social Media Icons: Clickable icons linking to Forever Modest's Instagram, Facebook, TikTok, and YouTube accounts

Email Subscription Field: Optional form for users to enter their email and receive newsletters or offers

Minimalist Style: Maintains a clean look with muted colors and legible fonts.

Accessibility Focus: Includes screen-reader-friendly elements and high contrast for readability.

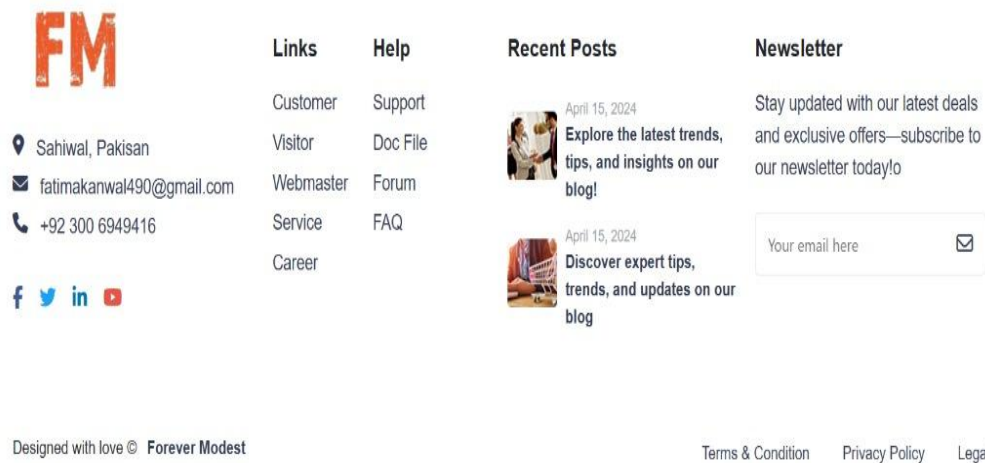


Figure 5.6: Footer Section of the Website

6.3.6. Contact Us Page

Provides a form for users to submit inquiries, feedback, or support requests. It includes fields for name, email, message, and a submit button.

interface allows users to directly communicate with the Forever Modest support team. It ensures easy and effective submission of inquiries.

Main Design Features:

- **Contact Form**
 1. Name Field
 2. Email Field
 3. Message Textbox
 4. Submit Button
- **Validation Rules:** Ensures all required fields are filled and email format is correct.
- **Support Information:** May include phone number, email address, or office hours.
- **Map Integration:** Optional map section showing office/store location (if applicable).
- **Feedback Confirmation:** After submission, user receives a message indicating successful inquiry or support request.

Contact Us

Stay Connected

Stay connected with us! Follow us for the latest updates, deals, and more."

**Address**
20, Mountain Street, Pakistan

**Call us**
+92 300 6949416

**Email Us**
hello@Forever Modest.com

Write your message here



Figure 5.7 Contact Us Page Interface

6.3.7. User Login

Allows users to sign in to their accounts securely. The page features fields for email and password, along with options for password recovery and new user registration. The login page allows users to access their personal accounts securely. It features modern design and security practices.

Main Design Features:

1. Input Fields

- Email
- Password

2. **Login Button:** Prominent and easily accessible.

3. **Forgot Password Link:** Redirects to password recovery workflow.

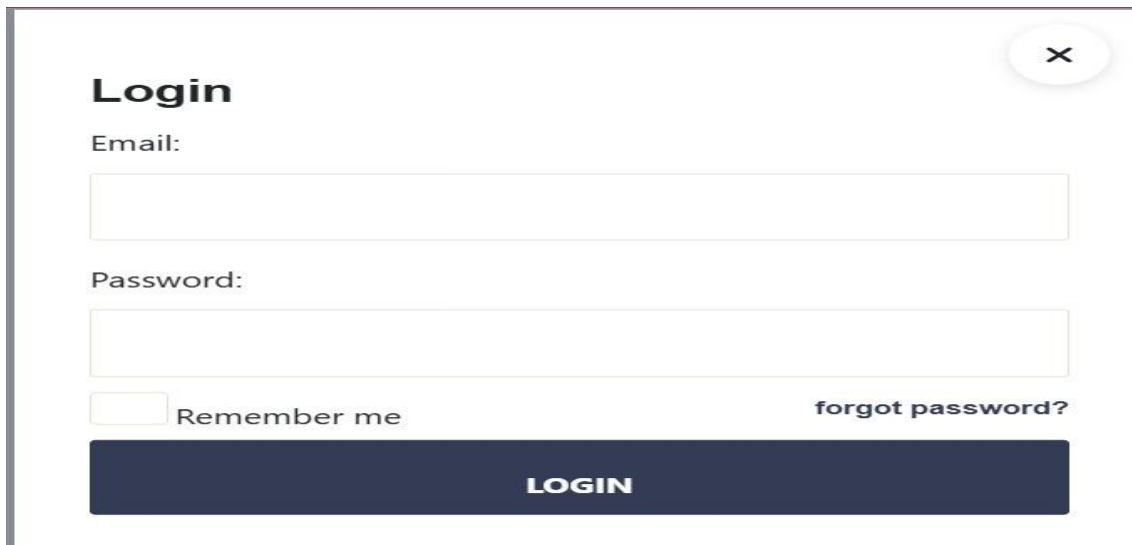
4. **Create Account Option:** Link for new users to sign up.

5. Security Emphasis:

- Backend validation of credentials

- Error handling for wrong email/password
- Captcha integration for both prevention (optional)

This screen ensures that users can log in confidently and proceed with shopping or checking their order history.



The image shows a user login interface. It features a dark blue header bar with a white 'x' icon in the top right corner. Below the header, the word 'Login' is displayed in a bold, dark font. The form consists of two input fields: 'Email:' and 'Password:', each with a corresponding text box. Below the password field, there is a checkbox labeled 'Remember me' and a link that says 'forgot password?'. At the bottom of the form is a large, dark blue button with the word 'LOGIN' in white capital letters.

Figure 5.8 User Login Interface

6.3.8. User Signup

The Sign Up page is a critical entry point for new users who want to create an account and access the features or services of an application or website. This page must be user-friendly, clear, and secure to ensure a smooth registration process that encourages users to complete it without frustration.

Purpose and Functionality:

The page collects four main pieces of information:

- **Username:** A unique identifier that the user will use on the platform.
- **Email Address:** Used for communication, verification, and password recovery.
- **Password:** The secret key that protects the user's account.
- **Confirm Password:** Ensures the user typed the intended password correctly by asking for reentry

➤ User Interface and Design Elements

1. Layout

The form is vertically aligned with labeled fields stacked for easy scanning and completion.

- **Labels:** Each input field has a clear label above it, which helps users understand what information is required.

- **Input Fields:** Each field has ample padding and spacing to prevent errors in data entry and to create a comfortable touch target, especially on mobile devices

2. Sign Up Button

Bold, large button that stands out visually, encouraging users to complete the sign-up process. The button is disabled or becomes active only when all fields are filled correctly (can be implemented for better UX).

3. Close Button (X)

Located at the top-right corner, this allows users to easily close the form without signing up, offering an escape route if they change their mind.

4. Visual Style:

The form uses a minimalistic and clean style with subtle borders and gentle shadows to maintain focus on the input areas without distractions.

➤ User Experience (UX)

1. The flow is straightforward, with no unnecessary steps or complicated input requirements, minimizing barriers to entry.
2. Immediate visual feedback (via borders or error messages, not shown here but can be implemented) informs users if they make mistakes.
3. The confirm password field reduces the risk of users accidentally entering a wrong password.
4. Mobile-friendly layout considerations (though not shown explicitly) ensure users on all devices can easily interact with the form.

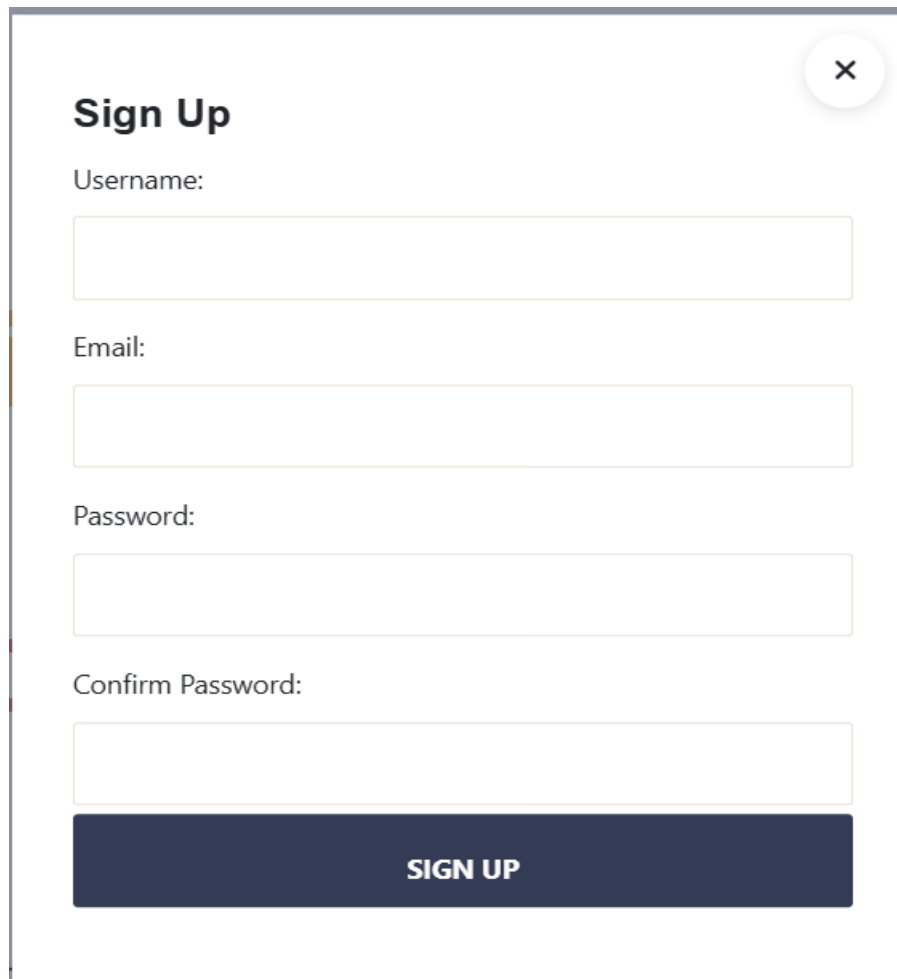
A user sign-up form with a dark blue header bar containing a close button (X). The form is titled "Sign Up" in bold. It includes four input fields: "Username:", "Email:", "Password:", and "Confirm Password:". Each field is a simple white rectangle with a thin border. Below the "Confirm Password" field is a dark blue button with the text "SIGN UP" in white, uppercase letters.

Figure 5.9 user signup interface

6.3.9. Blog Section Overview

- **Title & Intro**
The section is titled "**From Our Blog**" with a small heading "**Latest News**" above it, inviting users to check out recent updates or articles. The subtitles encourage users by saying they can find everything they need quickly.
- **Content Layout**
There are **three blog post previews** displayed side-by-side in a horizontal row.
- **Blog Post Previews**
Each preview includes:
 - An eye-catching **image** representing the blog content.
 - The **date** of the post (May 3, 2021).

A short **headline/title** of the blog post: "Lets make the begging to save the world again" (note: there's a typo — "begging" should likely be "beginning").

- **Navigation Dots**

Below the previews, there are **pagination dots** indicating there are more posts users can browse through, with the first dot highlighted, showing the current slide.

- **Additional UI**

There is an orange circular button with an arrow at the bottom-right corner, likely a **scroll-to-top** button for easy navigation.

This section effectively showcases blog content to engage visitors and encourage them to read the latest posts.

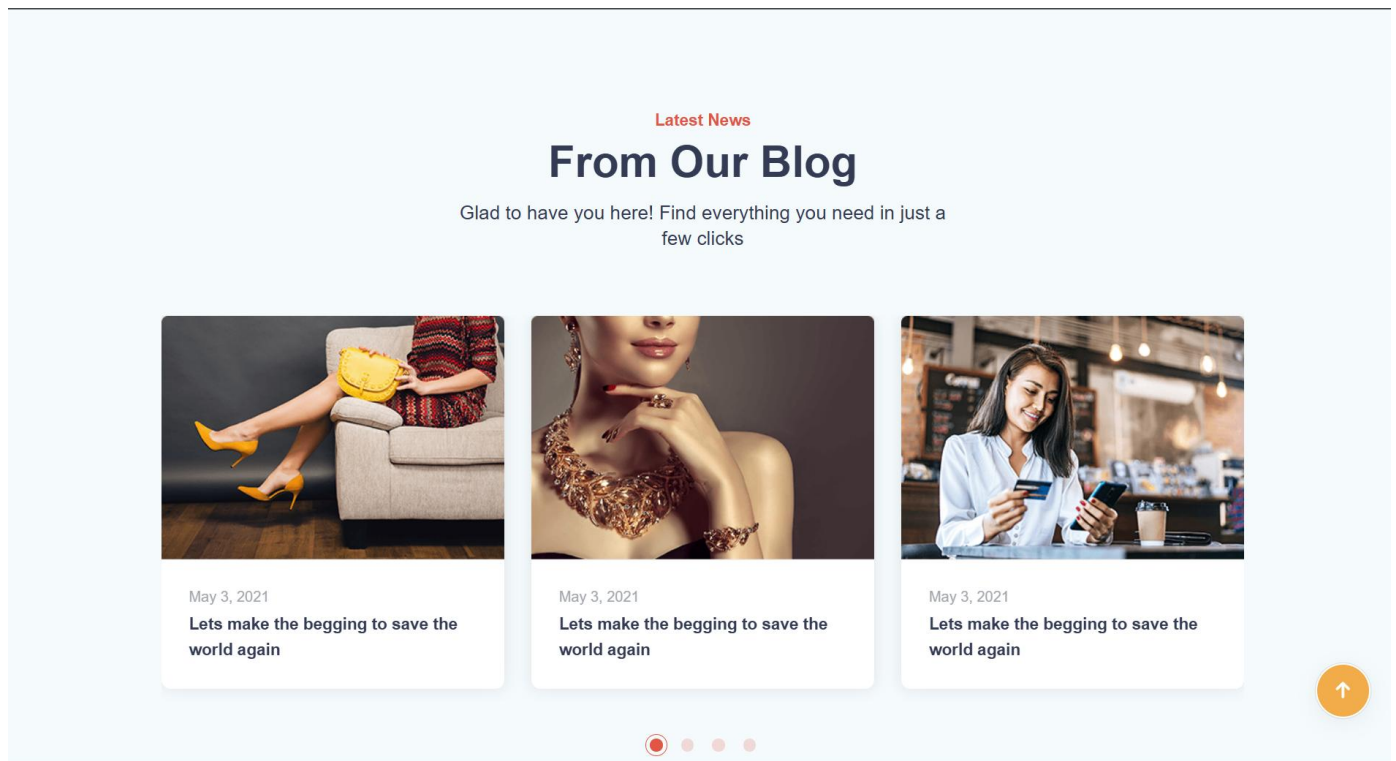


Figure 5.10 blog section overview

6.3.10. Top Navigation Bar

1. Menu Items

- HOME
- SHOP (with dropdown)
- COLLECTIONS (with dropdown)
- CONTACT

2. Help & Support A help/support option is shown on the right with a headset icon for customer assistance.

The navigation bar has a gradient red-orange background with white text.

➤ **Product Category Sections**

Below the navigation, there are four distinct clickable sections, each representing a product category:

1. Men's Section

- Image of a man holding shopping bags.
- Label below the image: "Men's Section" in red text.

2. Women's Section

- Image of a woman joyfully holding shopping bags against a red background.
- Label: "Women's Section."

3. Kid's Section

- Image of three kids standing together.
- Label: "Kid's Section."

4. Accessories Section

- Image displaying various accessories like sunglasses, jewelry, makeup, and a handbag.
- Label: "Accessories Section."

➤ **Purpose**

- This layout is used for easy navigation and category selection on a shopping site.
- Users can quickly jump to the section of products they want to browse or buy.
- Visual images help users identify categories quickly.

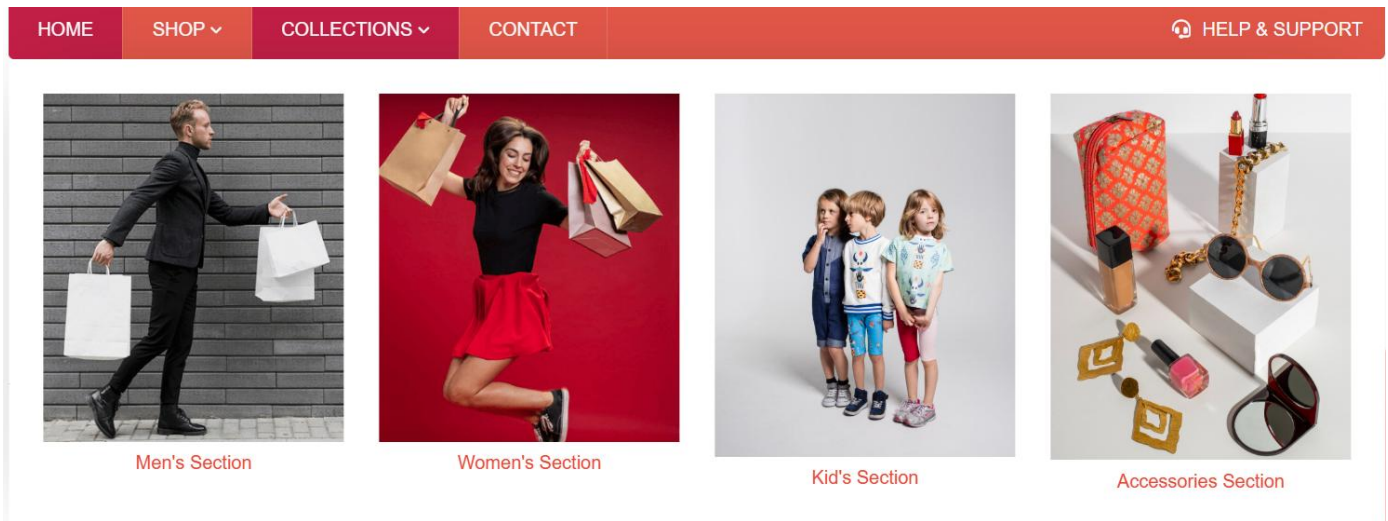


Figure 5.11 Top navigation bar

6.3.11. Adding New Products Page

The Add New Products page is a dedicated interface where the website administrator or seller can input details and upload information about new products to be listed on the online store. This page serves as a form to collect all the necessary data to create a complete product entry in the store's database.

Common Features and Sections

1. Product Information Form

- Product Name: Text input for the product title.
- Category Selection: Dropdown menu or multi-select to categorize the product (e.g., Men's, Women's, Accessories).
- Description: A text area or rich text editor for detailed product description.
- Price: Input for the product's price.
- Discount/Offer: Optional field for discount percentage or special offers.
- SKU (Stock Keeping Unit): Unique code for inventory management.

2. Product Images

- Upload functionality to add multiple images of the product.
- Image preview before uploading.
- Option to select a featured image.

3. Shipping Details

- Weight and dimensions for shipping calculation.
- Shipping options or restrictions.

4. Buttons

- **Save/Add Product:** To submit the form and save the new product to the database.
- **Cancel:** To discard changes and return to the product list or dashboard.

➤ Purpose

- To allow sellers or site admins to efficiently add new products.
- Ensure all relevant product information is collected consistently.
- Provide control over product display and inventory management.
- Support smooth updating and launching of new items on the store.

The screenshot shows the 'Add New Product' form on the FM website. The form is located below a navigation bar that includes a search bar, 'LOG IN', 'SIGN UP', and a shopping cart icon. The form itself has a white background and a light gray border. It contains several input fields: 'Product Name' (with a placeholder 'Enter product name'), 'Price' (with a placeholder 'Enter product price'), 'Description' (with a placeholder 'Enter product description'), and 'Category' (a dropdown menu with 'Select Category' as the selected option). There is also a 'Product Image' section with a 'Choose File' button and a 'No file chosen' status. An 'Add Product' button is located at the bottom right of the form.

Figure 5.12 adding new product catalog

6.3.12. Added Products in collection

Once a product is added through the Add New Products page by the admin or seller, it immediately becomes available and visible on the public-facing website product pages. This allows customers and visitors to browse, view details, and purchase the newly added items.

➤ Key Points:

- **Automatic Listing:** The newly added product automatically appears in the relevant product category or section (e.g., Men's Section, Women's Section, Kid's Section, Accessories Section) based on the category selected during product entry.
- **Product Display:** Each product is shown with its:
 - **Image(s):** The featured image and additional gallery images to attract customers.
 - **Name/Title:** The product name as entered in the form.
 - **Price:** Displayed clearly, including any discounts or offers.
 - **Short Description:** A brief summary or key features may be shown.

- **Detailed Product Page:** Customers can click on the product to view a full detailed page containing all product information, including extended descriptions, specifications, customer reviews (if enabled), and options to select size, color, or quantity.
- **Real-time Updates:** Any changes made to the product information by the admin, such as price updates, stock status, or images, will reflect instantly or shortly after on the website.
- **Search & Filter:** Products are searchable and filterable based on attributes like category, price, popularity, or other custom filters configured on the website.

➤ **Purpose and Benefit:**

This dynamic product display ensures that the website is always up-to-date with the latest products, providing seamless shopping experience to customers and helping the business keep the product catalog fresh and relevant

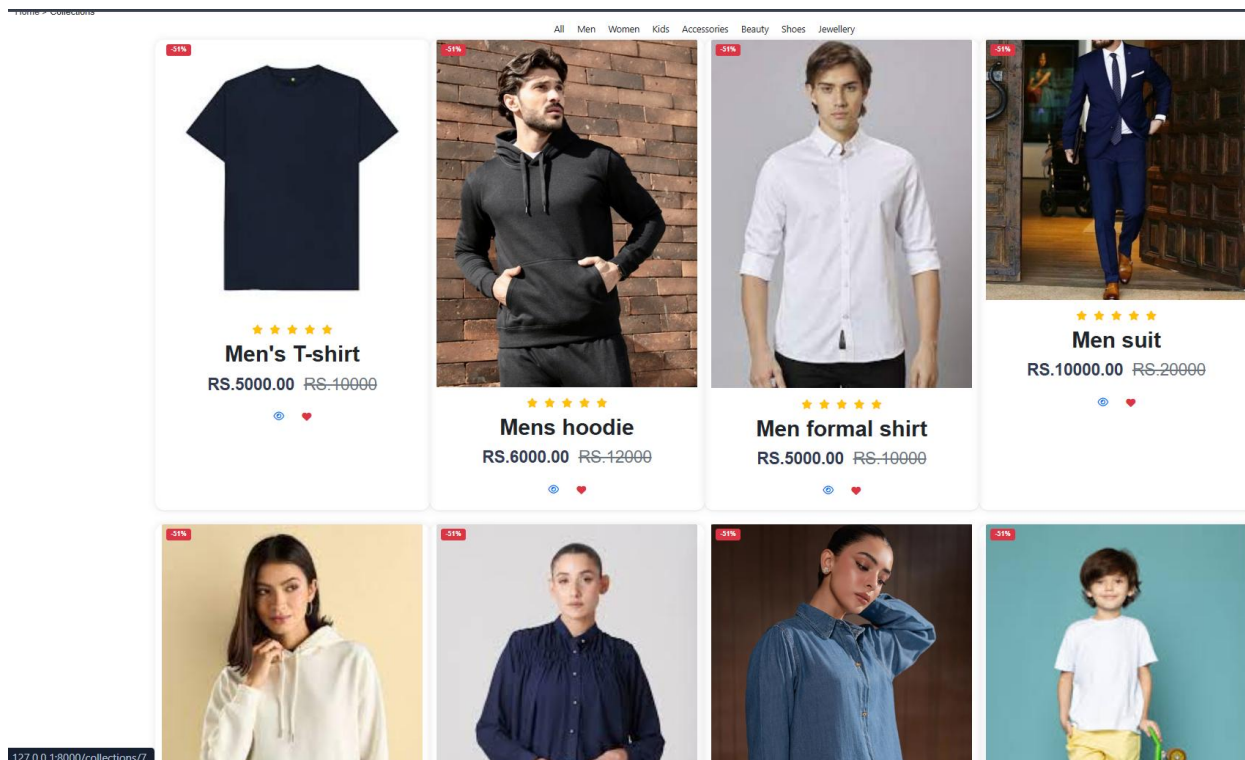


Figure 5.13 added products in collection

6.3.13. Add to Cart

The Add to Cart feature allows customers to select products they wish to purchase and temporarily store them in a virtual shopping cart before proceeding to checkout. This functionality enhances the shopping experience by enabling users to:

- **Select Multiple Products:** Customers can add one or more items to their cart without immediately purchasing them, allowing them to continue browsing.

- **Choose Quantity and Variants:** Users can specify the desired quantity and product options such as size, color, or model before adding the item to the cart.
- **View Cart Contents:** The cart displays a summary of all selected products, including product names, quantities, individual prices, and the total amount
- **Modify Cart:** Customers can update quantities, remove items, or save products for later, giving full control over their order before checkout.
- **Seamless Checkout Transition:** Once satisfied with their selections, users can proceed from the cart to the checkout page to complete their purchase.

Benefits

Simplifies the purchasing process by letting customers gather all desired items in one place. Provides flexibility to review and adjust the order. Improves customer satisfaction by making the shopping experience convenient and user-friendly.

Home > Collections > Mens hoodie



Mens hoodie

★★★★★

RS.6000.00 RS.12000

Men's Hoodie – Comfort Meets Style Stay warm, look cool. This men's hoodie is the perfect blend of comfort, functionality, and street-smart style. Crafted from a premium blend of cotton and polyester, it provides a soft, breathable feel while offering excellent insulation during colder days.

Key Features: Material: Soft-touch cotton blend for all-day comfort Fit: Regular fit with ribbed cuffs and hem for a snug feel Design: Adjustable drawstring hood and spacious kangaroo pocket Style: Minimalist look that suits casual outings, workouts, or lounging Durability: Fade-resistant fabric and strong stitching for long-lasting use

Whether you're layering up for a brisk morning or heading out with friends, this hoodie brings effortless style and warmth to your wardrobe. Available in multiple colors and sizes to match your personal style.

 Add to Cart



Figure 5.14 add to cart product

6.3.14. Checkout Page

"Forever Modest" originated from the observed limitations of physical stores and the need to move towards a digital platform that allows for increased accessibility and reach. In the modern world, customers demand convenience, variety, and speed. Traditional shops lack 24/7 access, quick browsing, or search capabilities. This project aims to resolve such issues by creating a

centralized web-based platform, where users can shop from anywhere and anytime.

Key features of your checkout process might include:

- **Order Summary:** Displays all products added to the cart, including quantity, price, and total cost, allowing users to verify their purchase before finalizing.
- **User Information Form:** Collects essential details such as name, shipping address, email, and phone number to process the order and delivery.
- **Payment Options:** Offers multiple payment methods (credit/debit cards, digital wallets, etc.) ensuring a secure and convenient transaction.
- **Review and Confirm:** Users can review all information and confirm their order with a final click.
- **Order Confirmation:** After successful payment, users receive an order confirmation message or email with details of their purchase and estimated delivery time.

Benefits

- Provides a clear and organized summary to avoid mistakes.
- Ensures secure payment processing to protect customer data.
- Enhances customer trust and satisfaction by offering a smooth checkout experience.
- Facilitates order tracking and communication through confirmation notifications.

> Checkout

Returning Customer?[Click Here to Login](#)

Your Billing Address

First Name *

Last Name

Phone *

Email Address

Country *

Select your Province

Address *

City / Town *

Zip Code *

Place Order

Coupon Code

Enter coupon code

[Apply Coupon](#)

Your Order

Item	Price
Mens hoodie	Rs. 6,000.00
Subtotal	Rs. 6,000.00
Shipping (Standard)	Rs. 20.00
Total	Rs. 6,020.00

Payment Method

☒ Cash On Delivery

Pay on delivery

Figure 5.15 Checkout Pa

Chapter 7

Testing and Evaluation

7. Testing and Evaluation

In any software development lifecycle, testing and evaluation play a pivotal role in ensuring that the final product meets the desired quality standards and fulfills the users' needs and expectations. This chapter presents a detailed overview of the testing and evaluation phases conducted for the e-commerce website project. The objective of testing is to systematically identify and rectify any defects, bugs, or logical errors in the system before the website is deployed to end users. This proactive approach minimizes post-deployment issues, enhances user satisfaction, and ensures robust system performance.

The testing process adopted for this project encompasses both manual testing and automated testing techniques, complemented by various testing strategies such as unit testing, functional testing, integration testing, and system testing. Each of these testing types targets different aspects of the system, providing a multi-layered verification to improve overall software quality.

7.1. Manual Testing

Manual testing was conducted to verify that each feature of the website works correctly by executing test cases without the use of automation tools. Manual testing is a traditional and fundamental testing approach where testers manually execute test cases without the assistance of automation tools. It requires careful preparation and execution of test scenarios to verify that the system behaves as expected under various conditions.

Objectives of Manual Testing

- To validate the correctness and completeness of individual features.
- To identify any usability or interface issues.
- To verify error handling and feedback mechanisms from a user's perspective.
- To check system behavior with both valid and invalid inputs.

Manual Testing Process in This Project

The manual testing phase in this e-commerce project involved a detailed review of key functionalities such as user registration, login, profile management, product browsing, cart operations, and checkout processes. Each feature was tested through a series of test cases designed to cover typical user workflows and edge cases.

For example, during the login process, testers manually input both valid and invalid credentials to verify if the system correctly authenticates users or provides appropriate error messages. Similarly, profile editing was tested by changing user details and observing whether changes were saved or if validation errors appeared for incorrect inputs.

Manual testing plays a crucial role in uncovering issues that may not be easily automated, such as visual inconsistencies, navigation difficulties, or ambiguous instructions, thus contributing to a more user-friendly design.

7.2. System testing

System testing was conducted after the development phase to ensure that all modules and features worked as expected. This involved checking the main functionalities such as user login, registration, profile editing, product listing, and cart management. The testing ensured that the requirements outlined during the analysis phase were successfully implemented. Three types of testing were performed: unit testing, functional testing, and integration testing. System testing is conducted after all development activities are complete to validate the entire integrated system against the specified requirements. Unlike unit or module-level tests, system testing considers the software as a whole, ensuring that all modules interact seamlessly to deliver the expected functionality.

Purpose of System Testing

- To confirm that all individual components work together harmoniously.
- To verify end-to-end business processes from the user's viewpoint.
- To validate the system's compliance with functional and non-functional requirements.
- To uncover defects that emerge from the integration of modules.

Key Functionalities Tested

In this project, system testing focused on the core e-commerce functionalities:

User Login and Registration: Ensuring users can create accounts, log in, log out, and recover passwords securely.

- **Profile Editing:** Allowing users to update their personal information accurately.
- **Product Listing:** Displaying products correctly with relevant details like name, price, and images.
- **Cart Management:** Adding, updating, and removing products from the shopping cart.
- **Checkout Process:** Completing purchases with valid payment and shipping information.

System testing also involved stress testing to check the system's behavior under high loads and usability testing to assess how easy and intuitive the website is for end-users.

7.3. Unit Testing

Unit Testing 1: User Login

Testing Objective: To verify that the login form functions correctly.

Table 6.1 To verify that the login form functions correctly

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Attempt to login with valid credentials	Username: user001 Password: pass123	Successfully logged in and redirected to the homepage	Pass
2	Attempt login with invalid password	Username: user001 Password:	Error message: "Invalid credentials" displayed	Pass

		wrong pass		
--	--	------------	--	--

System testing also involved stress testing to check the system's behavior under high loads and usability testing to assess how easy and intuitive the website is for end-users.

Unit testing is the foundational testing level focused on verifying the smallest testable parts of an application, typically individual functions or methods, to ensure they work correctly in isolation.

Importance of Unit Testing

- Detects defects early in the development process.
- Simplifies debugging by isolating issues to specific units.
- Ensures that code changes do not break existing functionality (regression testing).
- Provides documentation of expected behavior for developers.

Unit Testing in the Project

Two primary units were extensively tested:

1. User Login

Testing the login functionality involved verifying that the system accepts correct credentials and rejects incorrect ones, thereby enforcing access control.

- **Test Case 1:** Logging in with valid username and password should successfully grant access and redirect the user to the homepage.
- **Test Case 2:** Using an incorrect password should display an error message like "Invalid credentials," preventing access.

The results showed the login form behaved correctly under these conditions, providing confidence in the authentication mechanism.

2. Edit Profile

The profile editing feature was tested to ensure user inputs were correctly updated and validated.

- **Test Case 1:** Updating an email address with a valid email format should save successfully.
- **Test Case 2:** Leaving required fields, such as the user's name, empty should trigger an appropriate error message.

Successful completion of these tests demonstrated robust validation and data update logic in the

profile management module.

Testing Objective: To ensure that the edit profile form works correctly. **Table 6.2 ensure that the edit profile form works correctly.**

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Update user email	New Email: mannalsaeed628@gmail.com	Profile updated successfully	Pass
2	Leave required fields empty and submit	Name field left blank	Error message: "Name is required" displayed	Pass

7.4. Functional Testing

Functional testing was carried out after unit testing to ensure that all modules worked according to their specifications. Each function was tested in isolation with valid and invalid inputs to determine correct behavior. Functional testing evaluates individual functions of the software to confirm that they operate according to the specified requirements. It tests the system's actions based on user inputs, examining how the system responds.

Focus Areas of Functional

Functional testing ensures that all system features work as intended when used by end-users. It also validates access control, error handling, data processing, and output generation. Updating an email address with a valid email format should save successfully.

Functional Testing Example: Login with Different Roles

- Since the website caters to different user roles, functional testing verified role-based redirection:
- Admin users logging in should be directed to an admin dashboard featuring administrative controls.
- Regular customers should be routed to a product browsing homepage with shopping capabilities.
- Both scenarios passed testing, confirming that role-based access control and user experience customization were implemented correctly

Testing

Objective: To confirm that users are redirected to the correct dashboard based on their role.

Table 6.3 To confirm that users are redirected to the correct dashboard

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Login as admin	Username: admin01 Password: admin123	Redirected to admin dashboard with admin-specific features	Pass
2	Login as customer	Username: user001 Password: pass123	Redirected to customer homepage with product listings	Pass

7.5. Integration Testing

Integration testing was performed to check how different modules interact with one another and whether data flows smoothly across modules. Integration testing verifies the interaction between different modules or components of the software to ensure they work together as expected.

Objectives of Integration Testing

- Detect interface defects between modules.
- Validate data flow and communication among components.
- Ensure that combined functionalities deliver correct end-to-end behavior.

Integration Testing Scenarios

Several critical workflows were tested to verify smooth module integration:

- **Login and Profile Update:** After successful login, users should be able to update their profiles without session interruptions or data inconsistencies.
- **Add to Cart and Checkout:** Products added to the cart should appear correctly in the checkout page, with accurate pricing calculations.
- **Search and Filter:** The search function combined with price filtering should return accurate and relevant results.

The success of these tests indicates that the different parts of the e-commerce system are well integrated and provide a seamless user experience.

Table 6.4 Integration Testing Results for Core Functionalities

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Login and update profile	Valid credentials + new email	Successfully redirected and profile updated	Pass

2	Add product to cart and proceed to checkout	Select product + click checkout	Product appears in checkout page with correct pricing	Pass
3	Search for a product and apply filters	Product keyword + price filter	Filtered results displayed accurately	Pass

7.6. Automated Testing

Automated testing was performed on selected functionalities using a simple tool like Selenium to ensure consistency and speed in repetitive tasks such as login and navigation checks. Automated testing utilizes software tools to run tests repetitively and efficiently, reducing human effort and minimizing the risk of human error in the testing process

Why Automated Testing?

- Speeds up regression testing.
- Enables frequent and consistent test execution.
- Frees manual testers to focus on exploratory and usability testing.
- Provides quick feedback during continuous integration and deployment.

Tools and Implementation For this project, **Selenium** was employed for automating browser-based tests. Selenium is an open-source framework that simulates user interactions with web pages, allowing thorough testing of front-end functionalities.

Automated Test Coverage

The automated tests focused on:

- **Login Page:** Ensuring login functionality works under various scenarios.
- **Search Feature:** Validating search inputs return the correct products.
- **Navigation Links:** Checking that menu links lead to their correct pages without errors.

Test scripts ran successfully with all results passing, proving the reliability of these core functionalities across multiple test runs.

Table 6.5 Automated Testing Results Using Selected Tools

Tool Name	Tool Description	Applied On	Results
Selenium	Open-source tool for automating web browsers	Login page, Search feature, Navigation links	Test scripts executed successfully; all results passed

Chapter 8

Conclusion And Future Work

8. Conclusion and Future Work

The e-commerce website was successfully implemented and tested using both manual and automated methods. Testing ensured that the website's core functionalities, such as login, product listing, and user account management, were fully operational. In the future, more advanced features such as payment gateway integration, email confirmation, user reviews, and product recommendations can be implemented to enhance the website's functionality and user experience.

8.1. Conclusion

In conclusion, the **Forever Modest** e-commerce website was successfully developed as a user-friendly platform aimed at delivering a seamless online shopping experience. The platform caters to a diverse audience by offering a wide range of modest fashion products, including clothing, accessories, and jewelry. Key functionalities were implemented to improve customer satisfaction, including user registration and login, product browsing and filtering, shopping cart management, and order placement.

The successful development and implementation of the Forever Modest e-commerce website marks a significant milestone in delivering a dedicated platform for modest fashion consumers. The primary goal was to create a seamless and intuitive online shopping experience tailored to meet the unique needs of this growing market segment, and this objective has been achieved through a carefully designed and tested system.

User-Friendly and Accessible Design

The website was built with a strong emphasis on user-friendliness and accessibility. The interface features a clean and attractive layout that allows users to effortlessly navigate through product categories, search for specific items, and filter results based on preferences such as price, size, or style. Clear product images, detailed descriptions, and easy-to-understand pricing information contribute to informed decision-making for shoppers. Accessibility considerations, including readable fonts, high-contrast color schemes, and responsive design, ensure that users on various devices and with differing abilities can comfortably use the site.

Comprehensive Feature Set for Customer Satisfaction

Key functionalities that enhance the user experience were successfully integrated into the platform:

- **User Registration and Login:** Secure authentication allows customers to create personal accounts, facilitating personalized experiences, order tracking, and account management.
- **Product Browsing and Filtering:** Users can explore a wide array of modest fashion products ranging from clothing and accessories to jewelry, with intuitive filtering options

to quickly narrow down choices.

- **Shopping Cart and Checkout:** The cart system supports adding multiple items, adjusting quantities, and removing products before proceeding to checkout, where orders can be placed seamlessly.
- **Order Management:** Both customers and administrators can view and manage order details, ensuring transparency and smooth transaction handling.

Backend Administration and Content Management

The admin panel was developed to empower the site administrators with full control over product listings, including adding new items, editing existing ones, and removing outdated stock. This ensures that the website content remains fresh and relevant, reflecting current inventory and promotional campaigns. The backend also provides tools for monitoring orders and user activity, supporting effective business management and decision-making.

Value-Added Services

The Forever Modest platform incorporates value-added services designed to build customer trust and encourage repeat purchases, including:

- **24-Hour Delivery:** Offering expedited shipping options to meet urgent customer demands.
- **Free Shipping:** Providing free delivery for orders over a specified amount, a popular incentive that can boost average order value.
- **Easy Return Policy:** Allowing hassle-free returns within a reasonable timeframe, which reassures customers and reduces purchase hesitation.

Meeting Primary Objectives

Through these features and careful testing—both manual and automated—the Forever Modest e-commerce website has met the primary objectives set at the start of development. It provides a solid foundation for a modern, scalable, and secure e-commerce platform tailored specifically to the modest fashion industry.

8.2. Future Work

As Forever Modest evolves, several future enhancements can be considered to elevate its capabilities and user experience. While the current system provides a strong baseline, e-commerce is an ever-evolving field, and continual enhancement is crucial to staying competitive and relevant. The following future improvements are envisioned to significantly enhance Forever Modest's capabilities, user experience, and market reach.

Payment Gateway Integration

One of the most critical enhancements is integrating reliable and secure payment gateways such as **PayPal**,

Stripe, or direct credit/debit card processing. The implementation of payment gateways would allow customers to complete purchases instantly and securely within the website, improving convenience and trust.

- **Security Considerations:** Integration must comply with PCI DSS (Payment Card Industry Data Security Standard) to protect sensitive financial information.
- **Multiple Payment Options:** Offering a variety of payment methods, including digital wallets and bank transfers, can cater to diverse user preferences and expand accessibility.
- **Fraud Prevention:** Additional layers of verification, such as 3D Secure, can reduce the risk of fraudulent transactions, protecting both customers and the business.

Order Tracking System

Adding a real-time order tracking feature would provide customers with visibility into their purchase status from order confirmation through to delivery.

- **Shipment Status Updates:** Integration with logistics partners' APIs can automatically update tracking information on the customer dashboard.
- **Estimated Delivery Time:** Providing dynamic estimates based on current shipping conditions helps manage customer expectations.
- **Notifications:** Automated alerts via email or SMS about shipment milestones (dispatch, in transit, out for delivery) enhance transparency and customer satisfaction.

Recommendation Engine

Personalization is key to modern e-commerce success. Implementing a recommendation engine based on user behavior, purchase history, and product similarities would create a more engaging shopping experience.

- **Collaborative Filtering:** Suggest products that users with similar tastes have purchased.
- **Content-Based Filtering:** Recommend items similar to those the user has viewed or added to their cart.
- **Upselling and Cross-Selling:** Present complementary or higher-value products to increase average order size.

Such data-driven recommendations not only improve the customer journey but can also drive higher sales and customer retention.

Email & SMS Notifications

Timely communication through email and SMS notifications is essential for maintaining customer engagement.

- **Order Confirmations:** Immediately inform users that their order has been received and is being processed.
- **Promotional Campaigns:** Regular newsletters and special offer announcements can

boost sales and customer loyalty.

- **Shipping Updates:** Keep customers informed about delivery progress.
- **Feedback Requests:** Post-purchase messages encouraging reviews and ratings can improve community trust and provide valuable insights.

Implementing these communications through automated workflows ensures consistency and efficiency.

Mobile App Version

With the increasing dominance of mobile devices for online shopping, developing a dedicated mobile app for Forever Modest would provide users with:

- **Improved Accessibility:** Native apps often offer better performance and usability than mobile websites.
- **Push Notifications:** Direct alerts about sales, promotions, and order updates.
- **Offline Access:** Some features can be accessible without an active internet connection.
- **Enhanced User Experience:** Mobile-specific design and interactions tailored for smaller screens.

Developing apps for both **iOS** and **Android** platforms will broaden reach and convenience for customers.

Multilingual and Currency Support

To extend Forever Modest's market beyond its current locale, implementing multilingual interfaces and multiple currency options is essential.

- **Language Localization:** Translating the interface and content to various languages will attract non-English speaking users and enhance their shopping experience.
- **Currency Conversion:** Automatically displaying prices in the user's local currency with real-time exchange rates facilitates international sales.
- **Legal and Tax Compliance:** Supporting different countries requires incorporating local tax rules and legal standards for e-commerce.
- This global expansion effort can significantly increase the platform's audience and revenue potential.

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